Software Requirements Specification (SRS)

Project Title: Campus-Cruise

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1. Introduction

1.1 Purpose

Transportation is an essential part of university life, as students travel to and from campus on a daily basis. However, many students face challenges such as high transportation costs, long waiting times, traffic congestion, limited availability of reliable transportation options, and concerns regarding personal safety. Although public transportation and commercial ride-sharing services are available, they are often expensive, inconsistent, and lack the trust that comes from traveling within a verified university community.

Campus-Cruise is a multi-platform ride-sharing system developed exclusively for students of United International University (UIU). The platform aims to create a secure, affordable, and community-driven transportation network by connecting students who own vehicles with students seeking rides along similar routes. Through a combination of web and mobile applications, the system enables users to post rides, search for available rides, request bookings, communicate through an integrated chat system, negotiate fares, make payments, track rides in real time, and access emergency support features.

The primary objective of Campus-Cruise is to improve the commuting experience of UIU students by reducing transportation costs, enhancing safety, minimizing travel inconvenience, and promoting collaboration among members of the university community. By restricting access to verified students only, the platform establishes a trusted environment where users can confidently share rides and interact with one another.

This Software Requirements Specification (SRS) document provides a comprehensive description of the Campus-Cruise system and its requirements. The document outlines the functional requirements, non-functional requirements, system constraints, operating environment, user interactions, security mechanisms, and performance expectations of the proposed platform. It serves as a formal reference for all stakeholders involved in the project, including developers, designers, testers, project supervisors, administrators, and future maintenance teams.

Furthermore, the SRS acts as a communication bridge between stakeholders and the development team. It ensures that all participants have a clear understanding of the project's objectives, features, and expected behavior before implementation begins. By documenting requirements in a structured manner, the project team can reduce ambiguity, improve development efficiency, and ensure that the final system meets the needs and expectations of its intended users.

1.2 Intended Audience

This Software Requirements Specification document is intended for multiple groups of stakeholders who will be involved in the planning, development, testing, deployment, operation, and maintenance of the Campus-Cruise platform. Each audience group can utilize this document according to its specific responsibilities and objectives.

Project Supervisor and Academic Evaluators

The project supervisor, course instructors, and academic evaluators can use this document to understand the project's goals, system architecture, feasibility, and overall scope. The document provides a clear overview of the proposed solution and demonstrates how the system addresses real-world transportation challenges faced by university students.

System Analysts and Software Engineers

System analysts and software engineers can use this SRS as a foundation for system design and development. The requirements specified in this document help translate business needs and user expectations into technical solutions. It provides detailed information regarding workflows, system interactions, data requirements, and operational constraints.

Developers and Designers

Frontend developers, backend developers, mobile application developers, and UI/UX designers can use this document as a development guideline. The requirements defined here help ensure consistency throughout implementation and provide a clear understanding of system functionalities and user experience expectations.

Testing and Quality Assurance Team

The testing team can utilize the requirements specified in this document to create test plans, test cases, and validation procedures. Functional and non-functional requirements serve as measurable criteria for evaluating whether the system performs as intended and meets quality standards.

System Administrators

Administrators are responsible for managing users, verifying accounts, monitoring platform activities, reviewing reports, and maintaining platform integrity. This document provides information regarding administrative privileges, system controls, and operational procedures necessary for effective management.

End Users (UIU Students)

UIU students represent the primary users of the Campus-Cruise platform. The SRS helps users understand the services, features, and benefits offered by the system. It also clarifies user responsibilities, platform limitations, and the expected workflow for participating in ride-sharing activities.

Future Maintenance and Enhancement Teams

Future development and maintenance teams can use this document as a reference when upgrading the platform, fixing issues, integrating new features, or improving existing functionalities. A well-documented SRS reduces maintenance complexity and helps preserve the original objectives of the project throughout its lifecycle.

1.3 Scope of the System

Campus-Cruise is a secure ride-sharing platform designed exclusively for students of United International University (UIU). The system provides both mobile and web-based access, enabling students to share transportation resources within a trusted academic community. By connecting student drivers with student riders, the platform seeks to reduce transportation expenses, improve commuting efficiency, and enhance overall travel safety.

The system allows students to register using their official UIU email addresses and upload their student identification cards for verification. Drivers are additionally required to submit valid driving license information. After administrative approval, users can access the platform and participate either as riders, drivers, or both.

Drivers can create ride listings by specifying pickup locations, destinations, schedules, available seats, and fare details. Riders can search for rides, view ride information, and submit booking requests. Unlike conventional ride-sharing platforms, ride requests must be approved by drivers before confirmation, creating an additional layer of safety and control.

The platform includes an intelligent ride-matching mechanism that considers route similarity and class schedules. Students can upload their academic routines, allowing the system to recommend rides that align with their regular commuting patterns. This feature minimizes manual searching and improves ride availability for frequent travelers.

Campus-Cruise also incorporates several communication and safety mechanisms. Riders and drivers can communicate through an integrated real-time chat system, negotiate ride fares, coordinate pickup locations, and receive ride-related notifications. Real-time GPS tracking allows users to monitor ride progress, while an Emergency SOS feature provides immediate assistance options during unexpected situations.

The platform supports multiple payment methods, including digital payment gateways such as bKash and Nagad, as well as direct cash payments between users. After completing a ride, both riders and drivers can provide ratings and reviews, helping maintain trust and accountability within the community.

Major Functional Goals

1. Student Verification and Secure Access

- Registration using official UIU email addresses.
- Student ID verification.
- Driving license verification for drivers.
- Administrative approval before account activation.

2. Ride Management

- Ride creation and publishing.
- Ride searching and filtering.
- Ride request and approval workflow.
- Seat availability management.

3. Smart Ride Matching

- Route-based ride recommendations.
- Routine and schedule-based ride suggestions.
- Automated matching between riders and drivers.

4. Communication and Coordination

- Real-time chat functionality.
- Fare negotiation between users.
- Ride notifications and updates.

5. Payment Management

- bKash integration.
- Nagad integration.
- Direct cash payment support.

6. Safety and Security

- Verified university-only community.
- Real-time GPS tracking.

- Emergency SOS functionality.
- User ratings and reviews.

7. Administrative Management

- User verification and approval.
- Complaint and report management.
- User suspension and banning.
- Platform monitoring and moderation.

System Boundaries

The Campus-Cruise system operates within the following boundaries:

- Access is restricted exclusively to verified UIU students.
- The platform facilitates ride-sharing coordination but does not own or operate transportation vehicles.
- Drivers are responsible for maintaining valid driving licenses and vehicle-related legal requirements.
- The system supports ride coordination, communication, and payment facilitation but does not guarantee transportation availability.
- Administrative functions are limited to verification, monitoring, moderation, and user management activities.
- External services such as GPS, mapping systems, email services, and payment gateways are required for complete functionality.

Within these boundaries, Campus-Cruise provides a reliable, secure, and efficient transportation-sharing ecosystem designed specifically for the United International University community.

2. System Overview

2.1 Product Perspective

Campus-Cruise is a multi-platform ride-sharing system developed exclusively for the students of United International University (UIU). The system aims to solve the transportation challenges faced by students by providing a secure, affordable, and community-driven ride-sharing environment. Unlike commercial ride-sharing services such as Uber or Pathao, Campus-Cruise operates within a closed university ecosystem where only verified students can participate as riders or drivers.

The platform consists of a mobile application and a web application for students, while administrative functionalities are managed through a dedicated web-based admin panel. The system allows verified student drivers to offer rides and verified student riders to request rides, creating a collaborative transportation network within the university community.

Campus-Cruise operates as an independent system but relies on several external services to provide complete functionality. These integrations enhance security, communication, location tracking, and payment processing capabilities.

Relationship with Other Systems

1. University Email Verification Service

The platform uses official UIU email accounts during registration to verify student identity and restrict access to legitimate university members.

2. GPS and Mapping Services

The system integrates with GPS and mapping technologies to provide route visualization, pickup and drop-off location selection, live ride tracking, and distance calculation functionalities.

3. Payment Gateway Services

Digital payment gateways such as bKash and Nagad are integrated into the platform to facilitate secure online transactions between riders and drivers. Cash payment remains available as an alternative payment option.

4. Notification Services

Email and push notification services are used to inform users about ride requests, booking confirmations, ride cancellations, approval status, payment updates, and system announcements.

5. Mobile and Web Platforms

The system is accessible through Android devices, iOS devices, and modern web browsers, ensuring convenience and accessibility for all users.

System Context

Campus-Cruise acts as the central platform that connects riders, drivers, administrators, payment systems, notification services, and location-based services. The platform manages ride creation, ride booking, communication, payment coordination, and user verification while external services provide supporting functionalities such as authentication, navigation, and financial transactions.

2.2 Product Functions

Campus-Cruise provides a comprehensive set of functionalities designed to improve the daily commuting experience of UIU students. The major functions of the system are described below.

User Registration and Verification

- Registration using official UIU email addresses.
- Email verification for authentication.
- Student ID card submission and verification.
- Driving license verification for drivers.
- Administrative approval before account activation.

User Authentication and Account Management

- Secure login and logout functionality.
- Password recovery and reset features.
- Profile management and account updates.
- Role management for Rider and Driver accounts.

Ride Posting and Management

- Drivers can create ride offers.
- Route selection and destination management.
- Ride scheduling based on date and time.
- Seat availability management.
- Ride modification and cancellation.

Ride Search and Booking

- Search rides based on route and destination.
- Filter rides according to schedule and preferences.
- View driver information and ride details.
- Submit ride booking requests.
- Receive booking confirmation or rejection.

Smart Ride Matching

- Automatic ride recommendations.
- Route-based matching.
- Routine-based matching using class schedules.
- Efficient rider-driver pairing.

Real-Time Communication

- In-app chat between riders and drivers.
- Pickup coordination.
- Fare negotiation and bargaining.

- Ride-related discussions and updates.

Payment Management

- Online payment using bKash.
- Online payment using Nagad.
- Cash payment option.
- Payment confirmation and transaction records.

Live Ride Tracking

- Real-time GPS tracking.
- Driver location monitoring.
- Ride progress updates.
- Estimated arrival information.

Ratings and Reviews

- Driver rating system.
- Rider rating system.
- Feedback submission after ride completion.
- Community trust building through reviews.

Emergency Safety Features

- Emergency SOS button.
- Immediate alert generation.
- Enhanced safety monitoring.
- Emergency contact support.

Administrative Functions

- User verification and approval.
- Driver verification and monitoring.
- Complaint and report handling.
- User suspension and banning.
- Platform monitoring and moderation.

2.3 User Classes and Characteristics

The Campus-Cruise platform consists of three primary user classes, each with specific responsibilities, permissions, and interactions within the system.

1. Rider

A rider is a verified UIU student who uses the platform to search for transportation and request rides.

Characteristics:

- Must possess a valid UIU email account.
- Must complete student verification.
- Can search for available rides.
- Can submit ride booking requests.
- Can communicate with drivers through chat.
- Can make payments using supported methods.
- Can track rides in real time.
- Can provide ratings and reviews.

2. Driver

A driver is a verified UIU student who owns or operates a vehicle and offers rides to other students.

Characteristics:

- Must possess a valid UIU email account.
- Must submit a valid driving license.
- Must complete administrative verification.
- Can create and manage ride listings.
- Can accept or reject booking requests.
- Can communicate with riders.
- Can receive payments.
- Can receive ratings and reviews.

3. Administrator

Administrators are responsible for maintaining system security, integrity, and overall platform operations.

Characteristics:

- Access platform through a dedicated web panel.
- Verify students and drivers.
- Review submitted documents.
- Manage reports and complaints.
- Suspend or ban users when necessary.
- Monitor platform activities.
- Maintain system security and compliance.

External Systems

Several external systems support the operation of Campus-Cruise.

Examples include:

- UIU Email Service
- GPS and Mapping Services
- Notification Services
- Payment Gateways (bKash and Nagad)

These systems interact with the platform but are not considered direct users.

2.4 Operating Environment

The Campus-Cruise platform is designed to operate efficiently across multiple devices and environments.

Hardware Requirements

Server Infrastructure:

- Minimum 4 GB RAM
- Multi-Core Processor
- 100 GB Storage Capacity
- Cloud Hosting Support

User Devices:

Mobile Devices

- Android smartphones (Android 8.0 or higher)
- iPhones (iOS 13 or higher)
- GPS-enabled devices

Desktop and Laptop Devices

- Windows 10 or later
- macOS
- Linux-based systems

Software Requirements

Frontend Technologies:

- Flutter / React Native (Mobile Application)
- HTML5

- CSS3
- JavaScript

Backend Technologies:

- Node.js
- Express.js
- RESTful APIs

Database Technologies:

- MySQL
- PostgreSQL
- Firebase (Optional)

Network Requirements

- Stable Internet Connection
- HTTPS Secure Communication
- GPS Connectivity
- Push Notification Support
- Continuous Server Availability

2.5 Design and Implementation Constraints

Several constraints influence the design and implementation of the Campus-Cruise platform.

Restricted Access

Only verified UIU students are permitted to access the system.

Administrative Approval Requirement

Users cannot access ride-sharing functionalities until their accounts have been reviewed and approved by administrators.

Dependence on GPS Services

Location tracking, route matching, and navigation features depend heavily on GPS availability and accuracy.

Dependence on External Payment Systems

Online transactions depend on the availability and reliability of external payment gateway providers.

Cross-Platform Compatibility

The system must function consistently across web browsers, Android devices, and iOS devices.

Security Requirements

The platform must implement secure authentication, encrypted communication, and protected user data storage.

Resource Constraints

The project may initially utilize academic or low-cost development resources and cloud infrastructure during deployment.

2.6 Assumptions and Dependencies

The successful operation of Campus-Cruise relies on several assumptions and external dependencies.

Assumptions

- All users possess valid UIU email accounts.
- Students provide authentic personal information during registration.
- Drivers submit valid driving license information.
- Users have access to smartphones or internet-enabled devices.
- Students maintain internet connectivity while using the platform.
- Users comply with university policies and community guidelines.

Dependencies

- UIU email services remain operational.
- GPS and mapping services function correctly.
- bKash and Nagad services remain available.
- Notification services operate without interruption.
- Cloud servers maintain high availability.
- Mobile operating systems continue supporting application requirements.

The reliability of these external services directly impacts the overall effectiveness and performance of the platform.

2.7 SDLC Approach

The Campus-Cruise project follows the Agile Software Development Life Cycle (SDLC) methodology. Agile development was selected because it supports iterative improvement, flexibility, continuous feedback, and rapid adaptation to changing user requirements.

Phase 1: Requirement Analysis

Information is collected through surveys, interviews, benchmarking, observations, and stakeholder discussions. Functional and non-functional requirements are identified and documented.

Phase 2: System Design

System architecture, database design, UML diagrams, workflows, and UI/UX prototypes are prepared according to the gathered requirements.

Phase 3: Development

Frontend, backend, database, API integrations, GPS services, chat modules, payment modules, and security mechanisms are implemented during this phase.

Phase 4: Testing

Unit testing, integration testing, system testing, security testing, and user acceptance testing are performed to ensure system reliability and quality.

Phase 5: Deployment

The completed application is deployed to cloud servers and made available through web and mobile platforms for UIU students.

Phase 6: Maintenance and Enhancement

After deployment, the system is continuously monitored. Bugs are resolved, security updates are applied, performance is improved, and new features are introduced based on user feedback and future requirements.

The Agile SDLC approach ensures that Campus-Cruise remains scalable, adaptable, and capable of meeting the evolving transportation needs of the UIU community.

3. Requirements Elicitation & Analysis

3.1 Introduction

Requirements Elicitation and Analysis is one of the most important phases of software development. The success of a software system largely depends on how accurately user needs and business requirements are identified and transformed into system functionalities. For the Campus-Cruise project, requirements were gathered through surveys, observations, benchmark analysis, stakeholder discussions, and feasibility studies.

The objective of this phase was to identify the transportation-related challenges faced by UIU students and determine how a university-exclusive ride-sharing platform could effectively address these challenges. The collected requirements helped define the scope, functionalities, constraints, and quality expectations of the proposed system.

The requirement analysis process ensured that the final system would not only solve existing commuting problems but also provide a secure, affordable, and efficient transportation experience for students.

3.2 Benchmark Analysis

Benchmark analysis was conducted to evaluate existing ride-sharing solutions and identify features that could be adapted to the Campus-Cruise platform.

Benchmark Products

The following ride-sharing platforms were selected for comparison:

- Uber
- Pathao
- Campus-Cruise (Proposed System)

Benchmark Comparison Table:

Feature	Uber	Pathao	Campus-Cruise
Target Audience	General Public	General Public	Verified UIU Students
University Email Verification	No	No	Yes
Student ID Verification	No	No	Yes
Driver License Verification	Yes	Yes	Yes
Ride Booking	Yes	Yes	Yes
Ride Approval Workflow	Limited	Limited	Yes
Smart Matching	Basic	Basic	Advanced
Routine-Based Matching	No	No	Yes
In-App Chat	Limited	Limited	Yes
Fare Negotiation	No	No	Yes

Feature	Uber	Pathao	Campus-Cruise
GPS Tracking	Yes	Yes	Yes
SOS Feature	Limited	Limited	Yes
Rating & Reviews	Yes	Yes	Yes
bKash/Nagad Payment	Partial	Yes	Yes
Cash Payment	Yes	Yes	Yes
Admin Verification	No	No	Yes
Student Community Focus	No	No	Yes

Benchmark Findings

The benchmark analysis revealed that existing ride-sharing platforms provide efficient transportation services but primarily target the general public. These systems lack features that specifically address the needs of university students. Campus-Cruise introduces student verification, routine-based matching, administrative monitoring, and enhanced safety mechanisms to create a trusted transportation ecosystem within the university community.

3.3 Requirements Analysis

Purpose of Requirement Gathering

The purpose of requirement gathering was to understand student transportation behavior, identify existing problems, and determine the features required for a successful ride-sharing platform.

Specific objectives included:

- Identifying transportation challenges.
- Understanding student safety concerns.
- Determining ride-sharing preferences.
- Evaluating payment expectations.
- Assessing willingness to participate as riders or drivers.
- Defining system functionality and constraints.

Scope of Requirement Gathering

The requirement gathering process focused on:

- Transportation habits.
- Cost-related issues.
- Scheduling requirements.
- Safety expectations.
- Communication needs.
- Verification mechanisms.
- Administrative requirements.
- Payment preferences.

3.4 Information Collected

Organizational Information

The following organizational information was collected:

- University transportation environment.
- Student commuting patterns.
- Campus accessibility issues.
- Transportation cost concerns.

User Information

Information collected from students included:

- Preferred transportation methods.
- Daily transportation expenses.
- Ride-sharing interests.
- Trust and verification expectations.
- Safety concerns.

Process Information

The following business processes were analyzed:

- User registration.
- Identity verification.
- Ride posting.
- Ride searching.
- Ride booking.
- Payment processing.
- Complaint management.
- Emergency support.

Technical Information

Technical requirements collected included:

- Mobile application requirements.
- Web platform requirements.
- GPS integration requirements.
- Payment gateway integration requirements.
- Security and privacy requirements.

3.5 Sources of Information

Requirements were gathered from both internal and external sources.

Internal Sources

1. Student Survey

A survey was conducted among UIU students to understand transportation habits and expectations.

2. Observation

Student commuting behavior and transportation difficulties were observed during regular university activities.

3. Team Discussions

Project team discussions were conducted to identify possible solutions and refine system requirements.

External Sources

4. Existing Ride-Sharing Systems

Uber and Pathao were analyzed to understand standard ride-sharing workflows and industry best practices.

5. Online Research

Articles, case studies, blogs, and technology resources were reviewed.

6. Academic References

Research papers and software engineering resources were consulted to improve requirement analysis quality.

3.6 Survey Methodology

A structured online questionnaire was distributed among UIU students to collect data regarding transportation habits, commuting challenges, safety concerns, and ride-sharing preferences.

Survey Objectives

The survey aimed to:

- Understand current transportation behavior.
- Identify transportation-related problems.
- Evaluate student interest in ride-sharing.
- Determine preferred system features.
- Assess trust and safety expectations.

Participants

The survey included students from different departments and academic years of United International University.

Data Collection Method

Data was collected using Google Forms and analyzed to identify trends and user requirements.

Limitations

- Limited sample size.
- Responses restricted to survey participants.
- User opinions may change over time.

3.7 Survey Results and Analysis

Transportation Method Analysis

The survey indicates that rickshaw is the most commonly used transportation method among respondents. This suggests a significant dependence on short-distance transportation services.

Transportation Cost Analysis

Transportation cost emerged as the most frequently reported commuting challenge. Students expressed interest in reducing transportation expenses through ride-sharing and cost-sharing mechanisms.

Safety Analysis

Students highlighted safety as a major concern, particularly when traveling with unfamiliar individuals. This finding supports the implementation of identity verification, ratings, reviews, GPS tracking, and SOS functionality.

Verification Analysis

Student ID verification was identified as the strongest trust factor. This result directly influenced the decision to make UIU email verification and student ID verification mandatory.

Ride-Sharing Interest Analysis

Most respondents showed positive interest in using a university-exclusive ride-sharing platform, indicating strong potential for user adoption.

Matching Preference Analysis

Students preferred ride recommendations based on schedule and time compatibility. As a result, smart routine-based matching was included as a core system feature.

Payment Preference Analysis

Students preferred flexible payment options, including digital payments and cash transactions. Therefore, the system supports bKash, Nagad, and Cash payment methods.

Survey Conclusion

Survey findings strongly support the implementation of Campus-Cruise and validate the proposed system objectives.

3.8 Requirement Classification

Functional Requirements

- User Registration
- Login and Authentication
- Verification Management
- Ride Posting

- Ride Searching
- Ride Booking
- Smart Matching
- Chat System
- Payment Processing
- GPS Tracking
- SOS Support
- Rating and Review
- Complaint Management

Non-Functional Requirements

- Security
- Reliability
- Availability
- Performance
- Scalability
- Maintainability
- Usability

Business Requirements

- Affordable Transportation
- Improved Safety
- Community Building
- Transportation Efficiency

User Requirements

- Easy Navigation
- Fast Ride Booking
- Secure Communication
- Emergency Support

3.9 Requirement Prioritization (MoSCoW Analysis)

Must Have Features

- Registration and Login
- Student Verification
- Ride Posting
- Ride Search
- Ride Booking
- Driver Approval
- Admin Dashboard

Should Have Features

- In-App Chat
- Online Payment
- Ratings and Reviews
- Complaint System

Could Have Features

- Ride History Analytics
- Personalized Recommendations
- Advanced Reports

Future Enhancements

- AI-Based Demand Prediction
- Predictive Ride Suggestions
- University Shuttle Integration

3.10 Result Table

Requirement Driver	Observation	Decision
Cost Reduction	Major student concern	Cost-sharing model
Safety	High priority	GPS + SOS
Trust	Verification required	Email + ID Verification
Communication	Coordination needed	In-App Chat
Payments	Flexible payment preferred	bKash, Nagad, Cash
Scheduling	Time matching preferred	Routine-Based Matching
Monitoring	Accountability required	Admin Dashboard

3.11 Gap Analysis

Current Situation	Required Situation	Campus-Cruise Solution
High transportation cost	Affordable commuting	Cost-sharing rides

Current Situation	Required Situation	Campus-Cruise Solution
Unknown travel partners	Trusted users	Verified student network
No ride coordination	Organized ride-sharing	Smart matching
Limited communication	Real-time communication	In-App Chat
No emergency support	Safe transportation	SOS Feature

Gap Analysis Conclusion

The analysis demonstrates a significant gap between existing transportation methods and student expectations. Campus-Cruise bridges this gap by introducing a secure, affordable, and technology-driven transportation ecosystem designed specifically for UIU students.

3.12 Feature List Fixation

User Features

- Registration and Login
- Profile Management
- Email Verification
- Student ID Verification

Driver Features

- Driver Verification
- Vehicle Information Management
- Ride Posting
- Seat Management
- Booking Approval

Rider Features

- Ride Search
- Ride Booking
- Ride Tracking
- Payment Management

Smart Features

- Route Matching
- Time Matching

- Routine-Based Suggestions

Communication Features

- In-App Chat
- Fare Negotiation

Safety Features

- GPS Tracking
- SOS Button
- Ratings and Reviews
- Complaint Management

Administrative Features

- User Verification
- Driver Verification
- User Suspension and Ban
- Ride Monitoring
- Complaint Handling
- Analytics and Reports

3.13 Chapter Summary

This chapter presented the complete requirement elicitation and analysis process for Campus-Cruise. Through benchmark analysis, surveys, observations, and stakeholder discussions, the project team identified the key transportation challenges faced by UIU students and translated them into system requirements. The findings confirmed the need for a university-exclusive ride-sharing platform and guided the selection of features that will be implemented in the final system.

4. Functional Requirements

4.1 User Registration and Login

Description

The system shall allow UIU students to create accounts using their official university email addresses. During registration, users must provide personal information and upload the required verification documents. The system will authenticate users and maintain secure access control mechanisms.

Inputs

- UIU Email Address
- Password
- Student Name
- Student ID Number
- Student ID Card Image
- Contact Information

Outputs

- Successful Registration Confirmation
- Verification Pending Status
- Login Success or Failure Message
- User Dashboard Access

Interactions

- Email Verification Service
- User Database
- Authentication Module
- Admin Verification System

Dependencies

- Active UIU Email Account
- Internet Connectivity
- Verification Services

4.2 Student Verification

Description

The system shall verify student identities using official university credentials. Students must upload their student identification cards, and the system administrator must review and approve submitted information before granting access to ride-sharing features.

Inputs

- Student ID Number
- Student ID Card Image
- User Profile Information

Outputs

- Verified Student Status
- Rejected Verification Notice

- Pending Verification Status

Interactions

- Admin Dashboard
- User Database
- Verification Records

Dependencies

- Valid Student Information
- Administrative Review

4.3 Driver Verification and Vehicle Management

Description

Students who wish to act as drivers must submit vehicle information and valid driving license details. Administrators will verify the information before approving driver privileges.

Inputs

- Driving License Information
- Vehicle Type
- Vehicle Registration Number
- Seat Capacity
- Vehicle Images (Optional)

Outputs

- Approved Driver Status
- Rejected Driver Status
- Updated Driver Profile

Interactions

- Driver Database
- Admin Verification Module

Dependencies

- Valid Driving License
- Administrative Approval

4.4 Profile Management

Description

The system shall allow users to view and update personal information, contact details, profile pictures, and transportation preferences.

Inputs

- Profile Information
- Contact Details
- Profile Photo
- Preferences

Outputs

- Updated User Profile
- Profile Change Confirmation

Interactions

- User Database
- Authentication System

Dependencies

- Verified User Account

4.5 Ride Posting

Description

Verified drivers shall be able to create ride offers by providing route information, travel schedules, available seats, and fare expectations. Posted rides will become visible to eligible riders.

Inputs

- Pickup Location
- Destination
- Travel Date
- Travel Time
- Available Seats
- Expected Fare

Outputs

- Published Ride Listing
- Ride Confirmation Message

Interactions

- Ride Database
- Matching Engine
- Notification Service

Dependencies

- Verified Driver Account

4.6 Ride Search and Filtering

Description

The system shall allow riders to search available rides using various criteria such as destination, pickup location, travel time, fare, and seat availability.

Inputs

- Pickup Location
- Destination
- Travel Date
- Travel Time
- Search Filters

Outputs

- Matched Ride List
- Ride Details

Interactions

- Ride Database
- Matching System
- GPS Service

Dependencies

- Active Ride Listings

4.7 Smart Ride Matching

Description

The system shall automatically recommend suitable rides based on route similarity, travel schedules, and academic routines uploaded by users.

Inputs

- User Route Information
- Travel Schedule
- Academic Routine
- Available Ride Data

Outputs

- Recommended Ride Suggestions
- Matching Score

Interactions

- Matching Engine
- Ride Database
- User Profiles

Dependencies

- Available Ride Data
- User Schedule Information

4.8 Ride Booking Request

Description

The system shall allow riders to submit booking requests for selected rides. Requests will remain pending until reviewed by the corresponding driver.

Inputs

- Ride ID
- Rider ID
- Requested Seats

Outputs

- Booking Request Confirmation
- Pending Booking Status

Interactions

- Ride Database
- Driver Notification Service

Dependencies

- Available Seats

4.9 Booking Approval and Seat Management

Description

Drivers shall be able to accept or reject ride requests. Upon approval, seat availability will automatically update.

Inputs

- Driver Decision
- Booking Request Information

Outputs

- Booking Confirmation
- Booking Rejection
- Updated Seat Count

Interactions

- Ride Database
- Notification System

Dependencies

- Driver Approval

4.10 In-App Chat and Fare Negotiation

Description

The system shall provide a real-time communication channel between riders and drivers. Users may coordinate pickup locations, discuss ride details, and negotiate transportation fares.

Inputs

- Text Messages
- Fare Proposals
- Ride Information

Outputs

- Message Delivery
- Negotiation Records

Interactions

- Chat Server
- Booking System

Dependencies

- Active Booking Request

4.11 Online Payment Gateway

Description

The system shall support digital payment methods including bKash and Nagad. Riders may also choose cash payment if agreed upon with the driver.

Inputs

- Payment Amount
- Payment Method
- Transaction Information

Outputs

- Payment Confirmation
- Payment Failure Notice
- Transaction Record

Interactions

- bKash API

- Nagad API
- Payment Database

Dependencies

- Active Booking
- Available Payment Service

4.12 Live GPS Tracking

Description

The system shall provide real-time location tracking during active rides. Both riders and drivers can monitor trip progress through GPS integration.

Inputs

- GPS Coordinates
- Ride Status

Outputs

- Live Location Updates
- Estimated Arrival Information

Interactions

- GPS Service
- Mapping API
- Ride Management System

Dependencies

- Internet Connectivity
- GPS Availability

4.13 Emergency SOS Feature

Description

The system shall provide an emergency SOS button that users can activate during unsafe situations. Emergency alerts will be sent along with location information.

Inputs

- SOS Trigger
- User Location
- Ride Information

Outputs

- Emergency Alert
- Location Information
- Incident Record

Interactions

- Admin Dashboard
- Notification Service
- GPS Service

Dependencies

- Active Internet Connection
- GPS Availability

4.14 Rating and Review System

Description

The system shall allow riders and drivers to evaluate each other after ride completion. Ratings and reviews will contribute to user reputation scores.

Inputs

- Rating Score
- Review Comments

Outputs

- Updated Rating
- Stored Review

Interactions

- User Database
- Review Management System

Dependencies

- Completed Ride

4.15 Complaint and Reporting System

Description

Users shall be able to report inappropriate behavior, safety issues, payment disputes, or other concerns directly to administrators.

Inputs

- Complaint Details
- Supporting Evidence
- Ride Information

Outputs

- Complaint Ticket
- Investigation Status

Interactions

- Admin Dashboard
- Complaint Database

Dependencies

- Registered User Account

4.16 Admin Dashboard

Description

The system shall provide a dedicated web-based administrative panel for monitoring users, verifying accounts, handling complaints, reviewing reports, and maintaining system integrity.

Inputs

- Verification Requests
- Complaint Reports
- User Information

Outputs

- Approval Decisions
- User Restrictions
- Administrative Reports

Interactions

- User Database
- Ride Database
- Complaint Management System

Dependencies

- Administrator Account

4.17 Functional Requirement Summary

The functional requirements described above define the core operations of the Campus-Cruise platform. These functionalities collectively support user verification, ride management, communication, payment processing, safety monitoring, and administrative control. Together they ensure that the platform provides a secure, efficient, and user-friendly ride-sharing experience exclusively for UIU students.

5. External & Non-Functional Requirements

5.1 User Interfaces

The user interface of Campus-Cruise is designed to provide a simple, intuitive, and user-friendly experience for all users. Since the platform will be used frequently by students for daily commuting activities, the interface must minimize complexity while ensuring quick access to important functionalities.

The system consists of three primary interfaces:

1. Rider Interface
2. Driver Interface
3. Admin Interface

Rider Interface

The Rider Interface will be available through both mobile and web platforms. The interface will allow users to search rides, book rides, communicate with drivers, track rides, and manage payments.

Main Screens

- Login and Registration Screen
- Verification Screen
- Home Dashboard
- Ride Search Screen
- Ride Details Screen
- Booking Management Screen
- Chat Interface
- Payment Interface
- Live Tracking Interface
- SOS Emergency Screen
- Rating and Review Screen
- Profile Management Screen

Driver Interface

The Driver Interface will allow verified drivers to post rides, manage bookings, communicate with riders, and monitor ride activities.

Main Screens

- Driver Dashboard
- Ride Posting Screen
- Booking Request Management Screen
- Chat Interface
- Payment Tracking Screen
- Ride History Screen
- Profile Management Screen

Admin Interface

The administrative interface will be web-based and accessible only to authorized administrators.

Main Screens

- Admin Login
- Verification Dashboard
- User Management
- Driver Management
- Ride Monitoring
- Complaint Management
- Payment Monitoring
- Analytics and Reporting

Navigation Design

The navigation structure will follow a simple hierarchical model.

Rider Navigation

Home → Search Ride → Ride Details → Booking → Payment → Tracking → Review

Driver Navigation

Dashboard → Post Ride → Booking Requests → Ride Management → Payment Tracking

Admin Navigation

Dashboard → Verification → Users → Rides → Complaints → Reports

User Interface Design Principles

The following design principles will be followed:

- Simplicity
- Consistency
- Accessibility
- Responsiveness
- Visibility of System Status
- Error Prevention
- User Feedback
- Mobile-First Design

Prototype Considerations

The prototype will include:

- Mobile Application Prototype for Riders and Drivers.
- Responsive Web Interface for Riders and Drivers.
- Dedicated Desktop Admin Panel.

The prototype screens will demonstrate the workflow of registration, verification, ride posting, ride searching, booking, payment processing, and emergency support.

5.2 Software Interfaces

Campus-Cruise interacts with several external software components and services to provide complete functionality.

Email Verification Service

The system will integrate with email services to verify UIU student email addresses during registration.

Purpose

- Identity Verification
- Account Activation
- Password Recovery

GPS and Mapping Services

The platform will integrate with mapping APIs to provide navigation and location-based functionalities.

Purpose

- Route Visualization
- Distance Calculation
- Pickup and Drop-Off Selection
- Live Tracking

Payment Gateway Services

The system will integrate with digital payment providers.

Supported Payment Methods

- bKash
- Nagad
- Cash Payment

Purpose

- Secure Transactions
- Payment Verification
- Transaction History

Notification Services

The system will send notifications for important events.

Notification Types

- Ride Request Notifications
- Booking Confirmations

- Ride Cancellations
- Payment Confirmations
- SOS Alerts
- Administrative Announcements

Database Interface

The platform will communicate with a centralized database system.

Stored Information

- User Information
- Ride Information
- Booking Records
- Payment Records
- Chat Messages
- Reviews
- Complaints

Authentication Services

Authentication services will handle secure login and user session management.

Functions

- User Authentication
- Session Management
- Password Encryption
- Access Control

5.3 Performance Requirements

The Campus-Cruise platform must maintain acceptable performance under normal and peak operating conditions.

Response Time Requirements

The system should provide fast responses to user actions.

Function	Expected Response Time
Login	Less than 3 Seconds
Ride Search	Less than 5 Seconds

Function	Expected Response Time
Booking Request	Less than 3 Seconds
Chat Message Delivery	Less than 2 Seconds
Payment Confirmation	Less than 10 Seconds
Ride Tracking Update	Real-Time

Scalability Requirements

The system should be capable of supporting future growth.

Requirements

- Support increasing numbers of users.
- Handle growing ride databases.
- Support future university expansion.
- Accommodate additional services.

Reliability Requirements

The system should maintain stable operation and minimize failures.

Requirements

- Accurate booking records.
- Consistent ride information.
- Reliable payment processing.
- Reliable notification delivery.

Availability Requirements

The platform should remain accessible whenever students need transportation services.

Requirements

- 24/7 Availability
- Minimal Downtime
- Continuous Database Access
- Continuous Server Monitoring

Capacity Requirements

The system should support:

- Multiple Concurrent Users
- Multiple Ride Requests
- Simultaneous Chat Sessions
- Real-Time Tracking Operations

Fault Tolerance

The system should recover gracefully from failures.

Examples

- Network Interruption Recovery
- Database Backup Restoration
- Session Recovery
- Payment Failure Handling

5.4 Security Requirements

Security is a critical aspect of Campus-Cruise because the platform manages personal information, location data, payment transactions, and user interactions.

Authentication Requirements

The system must ensure that only authorized users can access protected resources.

Security Measures

- Secure Login System
- Password Encryption
- Session Management
- Multi-Level Verification

Identity Verification

Users must complete identity verification before accessing ride-sharing services.

Verification Methods

- UIU Email Verification
- Student ID Verification
- Driver License Verification
- Administrative Approval

Access Control

Different users must have different access permissions.

Access Levels

Rider

- Search Rides
- Book Rides
- Make Payments
- Submit Reviews

Driver

- Post Rides
- Manage Bookings
- Receive Payments

Admin

- Verify Users
- Monitor Activities
- Manage Complaints
- Suspend Accounts

Data Protection

The system must protect sensitive information.

Protected Data

- Personal Information
- Student Identification Data
- Driving License Information
- Payment Information
- Location Information

Communication Security

All communication channels should use secure transmission mechanisms.

Security Techniques

- HTTPS Encryption
- Secure API Communication

- Token-Based Authentication
- Encrypted Password Storage

Security Monitoring

The system should continuously monitor suspicious activities.

Monitoring Activities

- Unauthorized Login Attempts
- Fake Verification Attempts
- Payment Fraud Detection
- Suspicious User Activities

Emergency Security Support

Emergency features should be protected against misuse.

Controls

- Verified User Requirement
- Ride-Based SOS Activation
- Admin Review of Emergency Incidents

5.5 Maintainability

The Campus-Cruise platform should be designed to facilitate future maintenance, modifications, and feature enhancements.

Modular Architecture

The system should follow a modular design approach.

Modules

- Authentication Module
- Ride Management Module
- Booking Module
- Payment Module
- Chat Module
- Tracking Module
- SOS Module
- Admin Module

A modular architecture allows individual components to be modified without affecting the entire system.

Code Maintainability

The software should follow coding standards and best practices.

Requirements

- Clean Code Structure
- Consistent Naming Conventions
- Proper Documentation
- Version Control Usage

Database Maintainability

The database should be structured to support future growth.

Requirements

- Normalized Database Design
- Efficient Indexing
- Backup Procedures
- Easy Data Migration

Extensibility

The system should support future enhancements.

Potential Future Features

- AI-Based Ride Recommendations
- Demand Prediction
- University Shuttle Integration
- Multi-University Expansion
- Subscription Services

Testing and Maintenance Support

The platform should support efficient testing and maintenance activities.

Requirements

- Unit Testing
- Integration Testing

- Automated Testing Support
- Error Logging
- Performance Monitoring

Documentation Requirements

Comprehensive documentation should be maintained to support future development teams.

Documentation Includes

- System Documentation
- API Documentation
- Database Documentation
- User Manuals
- Administrator Guides

5.6 Chapter Summary

This chapter described the external interfaces and non-functional requirements of the Campus-Cruise platform. The specified requirements ensure that the system remains secure, reliable, scalable, maintainable, and user-friendly. These quality attributes complement the functional requirements and establish the standards that the system must satisfy throughout its lifecycle.

6. Feasibility & Project Management Analysis

6.1 Introduction

Before developing the Campus-Cruise platform, a feasibility study was conducted to determine whether the proposed system could be successfully implemented within the available resources, technology, budget, and time constraints. The purpose of this analysis is to evaluate the practicality of the project and identify potential challenges that may arise during development and deployment.

The feasibility study examines the project from multiple perspectives, including technical, economic, operational, human resource, market, legal, and scheduling aspects. The results help determine whether Campus-Cruise is a viable solution for addressing the transportation challenges faced by UIU students.

6.2 Feasibility Analysis

6.2.1 Technical Feasibility

Technical feasibility evaluates whether the required technology, tools, and infrastructure are available to develop and operate the proposed system.

Campus-Cruise can be developed using modern software technologies that are widely available and well-supported. The project requires mobile application development, web development, backend services, database management, GPS integration, real-time communication, and payment gateway integration. All of these technologies are currently mature and accessible.

Required Technologies

- Mobile Application Development Framework (Flutter / React Native)
- Web Development Framework
- Backend API Services
- Cloud Database
- GPS and Mapping Services
- Payment Gateway APIs
- Authentication Services

Technical Advantages

- Availability of open-source development tools.
- Existing APIs for mapping and payments.
- Cloud hosting solutions are widely available.
- Modern smartphones support required functionalities.

Technical Challenges

- Real-time GPS tracking implementation.
- Payment gateway integration.
- Managing concurrent user requests.
- Ensuring secure communication.

Technical Feasibility Verdict

The project is technically feasible because all required technologies, tools, and development resources are available and can be implemented using standard software engineering practices.

6.2.2 Economic / Financial Feasibility

Economic feasibility evaluates whether the expected benefits justify the costs associated with development and operation.

Campus-Cruise is being developed primarily as an academic project and therefore requires relatively low initial investment. Most development tools, frameworks, and cloud services provide free or educational usage plans.

Expected Development Costs

Cost Component	Estimated Cost
Development Tools	Low
Hosting Services	Low
Database Services	Low
Testing Resources	Low
Maintenance	Moderate

Expected Benefits

- Reduced transportation costs for students.
- Improved transportation efficiency.
- Enhanced student safety.
- Better utilization of available vehicles.
- Reduced dependence on expensive ride-sharing services.

Financial Risks

- Future hosting costs may increase.
- Payment gateway transaction charges.
- Server maintenance expenses.

Economic Feasibility Verdict

The project is economically feasible because implementation costs are relatively low compared to the potential benefits provided to students.

6.2.3 Operational Feasibility

Operational feasibility determines whether the proposed system can function effectively within its intended environment.

The survey results indicate that students experience transportation challenges and are interested in a university-exclusive ride-sharing solution. The proposed platform aligns closely with user expectations and operational needs.

Operational Benefits

- Improved commuting convenience.
- Lower transportation costs.
- Increased trust through verification.
- Better ride coordination.
- Enhanced safety through tracking and SOS features.

Operational Challenges

- User adoption during the initial phase.
- Maintaining an adequate number of drivers.
- Handling user disputes.
- Monitoring misuse of the platform.

Operational Feasibility Verdict

The system is operationally feasible because it directly addresses existing transportation problems and has demonstrated user demand through survey responses.

6.2.4 Human Resource Feasibility

Human resource feasibility evaluates whether sufficient personnel and expertise are available to develop and maintain the system.

The project team possesses the necessary academic background in software engineering, database management, web development, mobile application development, and system analysis.

Required Roles

- System Analyst
- UI/UX Designer
- Mobile Application Developer
- Backend Developer
- Database Designer
- Tester
- Documentation Specialist

Available Resources

- Project Team Members
- Academic Supervisor
- Development Tools
- Educational Resources

Human Resource Challenges

- Balancing academic workload with development activities.
- Learning advanced technologies if required.
- Coordinating team activities.

Human Resource Feasibility Verdict

The project is feasible from a human resource perspective because the required knowledge and skills can be acquired and applied by the development team.

6.2.5 Market Feasibility

Market feasibility evaluates whether there is sufficient demand for the proposed solution.

The survey conducted among UIU students demonstrated significant interest in a campus-exclusive ride-sharing platform. Students identified transportation cost, convenience, and safety as major concerns, indicating a clear need for the proposed system.

Market Opportunities

- Large student population.
- Daily transportation demand.
- Increasing adoption of mobile applications.
- Growing interest in shared transportation services.

Market Risks

- Competition from Uber and Pathao.
- Limited driver participation.
- Resistance to adopting new technology.

Market Feasibility Verdict

The project is market feasible because it addresses a real problem experienced by a clearly defined user group.

6.2.6 Legal and Political Feasibility

Legal and political feasibility examines whether the system complies with applicable regulations and organizational policies.

Campus-Cruise operates within a university environment and focuses on connecting verified students rather than acting as a commercial transportation provider.

Legal Considerations

- User privacy protection.
- Data security compliance.
- Identity verification procedures.
- Secure payment handling.

Policy Considerations

- Compliance with university regulations.
- Appropriate user behavior policies.
- Anti-harassment measures.
- Fair usage guidelines.

Potential Legal Risks

- Data privacy violations.
- Misuse of personal information.
- User disputes.

Legal and Political Feasibility Verdict

The system is legally feasible provided that appropriate security, privacy, and user management policies are implemented.

6.2.7 Schedule Feasibility

Schedule feasibility evaluates whether the project can be completed within the available timeframe.

The development process is divided into multiple phases, allowing the project to be completed incrementally.

Project Timeline

Phase	Duration
Requirement Analysis	1–2 Weeks
System Design	1–2 Weeks

Phase	Duration
Prototype Development	3–4 Weeks
Admin Panel Development	2–3 Weeks
Advanced Features	2–3 Weeks
Testing	1–2 Weeks
Documentation	Continuous

Schedule Risks

- Requirement changes.
- Technical implementation challenges.
- Team availability constraints.

Schedule Feasibility Verdict

The project is schedule feasible because the proposed development plan can be completed within the available academic timeframe.

6.3 Overall Feasibility Summary

Feasibility Area	Verdict
Technical Feasibility	Feasible
Economic Feasibility	Feasible
Operational Feasibility	Feasible
Human Resource Feasibility	Feasible
Market Feasibility	Feasible
Legal & Political Feasibility	Conditionally Feasible
Schedule Feasibility	Feasible

The feasibility analysis indicates that Campus-Cruise is a practical and achievable project with strong potential to improve the commuting experience of UIU students.

6.4 SWOT Analysis

SWOT Analysis was conducted to identify the internal strengths and weaknesses of the project as well as external opportunities and threats.

6.4.1 Strengths

Verified Community

Only verified UIU students can access the platform, increasing trust and safety.

Cost Reduction

Students can share transportation expenses, reducing commuting costs.

Enhanced Safety

GPS tracking, SOS functionality, and user verification provide a safer travel environment.

Administrative Monitoring

The dedicated admin panel improves accountability and platform management.

Smart Matching

Route-based and routine-based matching improve ride availability and efficiency.

6.4.2 Weaknesses

Limited Initial User Base

The platform is restricted to UIU students, limiting the number of available riders and drivers.

Dependence on External Services

GPS, payment gateways, and notification services are controlled by third parties.

Driver Availability

The effectiveness of the platform depends on having enough active drivers.

Verification Complexity

Manual verification may increase administrative workload.

6.4.3 Opportunities

Expansion to Other Universities

The platform can be adapted for other educational institutions.

Additional Transportation Services

Future versions may include shuttle services and carpooling options.

AI-Based Recommendations

Artificial intelligence can be integrated for better ride suggestions and demand prediction.

Sustainability

Ride-sharing reduces traffic congestion and supports environmentally friendly transportation practices.

6.4.4 Threats

Competition

Existing ride-sharing platforms such as Uber and Pathao may attract potential users.

Low User Adoption

Students may initially hesitate to switch from familiar transportation options.

Security Incidents

Any safety-related incident could negatively affect user trust.

Technical Failures

GPS failures, payment issues, or server downtime could impact service quality.

6.5 SWOT Summary Table

Strengths	Weaknesses
Verified student community	Limited user base
Lower transportation cost	Driver dependency
GPS tracking and SOS	Verification workload
Administrative oversight	External service dependency
Smart matching system	Initial adoption challenges

Opportunities	Threats
Expansion to other universities	Competition from existing platforms
AI integration	Low user adoption
Sustainable transportation	Security incidents
Additional transportation services	Technical failures

6.6 Chapter Summary

This chapter evaluated the feasibility of the Campus-Cruise platform from technical, economic, operational, human resource, market, legal, and scheduling perspectives. The analysis indicates that the project is feasible and capable of addressing the transportation challenges faced by UIU students. Additionally, the SWOT analysis highlighted the strengths, weaknesses, opportunities, and threats associated with the project, providing valuable insights for future planning and risk management.

7. System Design (Diagrams)

7.1 Introduction

System design is a crucial phase of software development that transforms requirements into a structured technical solution. The purpose of system design is to define how the Campus-Cruise platform will function internally and how different components will interact with one another.

This chapter presents the major design diagrams used to model the system. These diagrams help developers, testers, stakeholders, and future maintenance teams understand the architecture, workflows, user interactions, and data movement within the platform.

The primary diagrams included in this chapter are:

- Data Flow Diagram (DFD)
- Use Case Diagram
- Class Diagram
- Sequence Diagram

These diagrams collectively provide a comprehensive representation of the Campus-Cruise system.

7.2 Data Flow Diagram (DFD)

7.2.1 Overview

The Data Flow Diagram (DFD) illustrates how information moves through the Campus-Cruise platform. It identifies the flow of data between users, system processes, databases, and external services.

The DFD helps visualize how ride requests, bookings, payments, communications, and administrative actions are processed within the system.

External Entities

The primary external entities involved in the system are:

- Rider
- Driver
- Administrator
- Payment Gateway
- GPS and Mapping Service

Major Processes

The main processes represented in the DFD include:

1. User Registration and Verification
2. Ride Management
3. Ride Booking
4. Payment Processing
5. Chat Communication
6. GPS Tracking
7. Complaint Handling
8. Emergency SOS Management

Data Stores

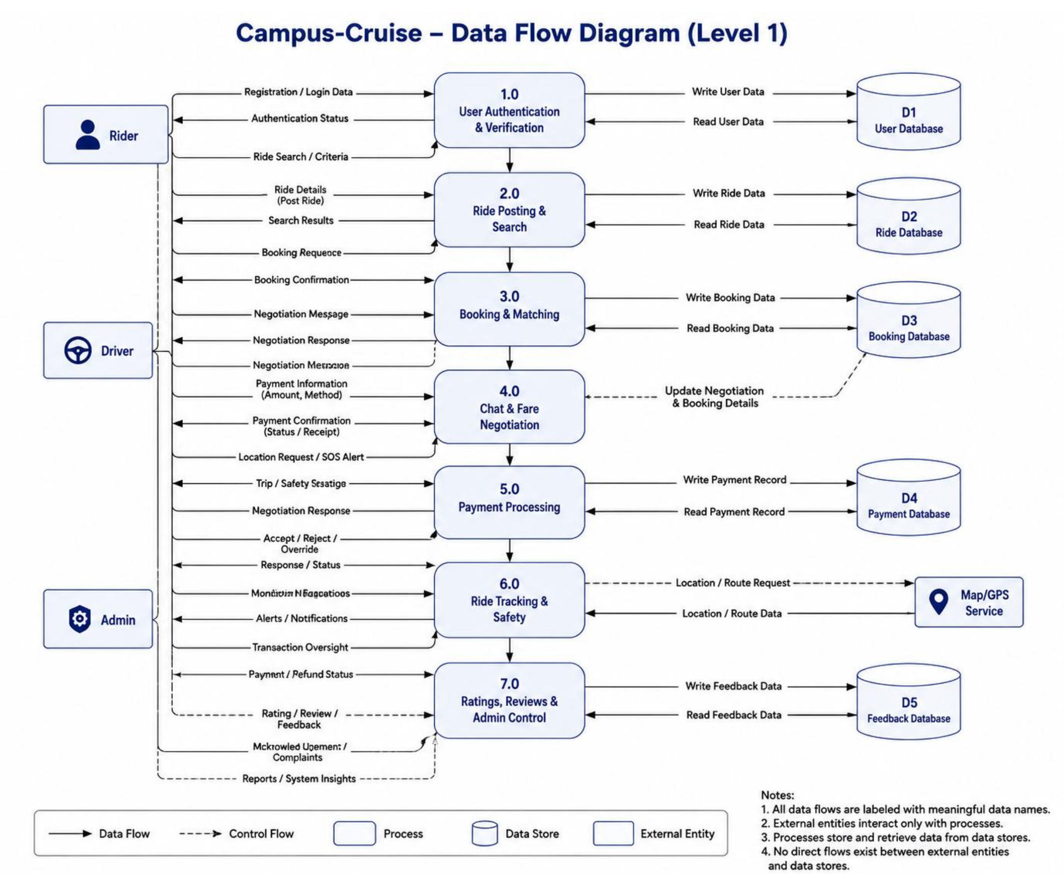
The DFD contains the following data stores:

- User Database
- Ride Database
- Booking Database
- Payment Database
- Message Database
- Review Database
- Complaint Database

DFD Description

The rider and driver interact with the system through registration, booking, ride management, and communication processes. User information is stored in the user database, while ride information is maintained in the ride database. Booking records, payment transactions, chat messages, reviews, and complaints are stored in their respective databases. The administrator interacts with the system through monitoring and management processes.

Figure 7.1: Data Flow Diagram



7.3 Use Case Diagram

7.3.1 Overview

The Use Case Diagram illustrates the interactions between system users and the Campus-Cruise platform. It provides a high-level representation of the functionalities available to each actor.

The diagram identifies the actions that riders, drivers, and administrators can perform within the system.

Actors

Rider

The rider is a verified UIU student who uses the platform to search and book rides.

Driver

The driver is a verified UIU student who provides transportation services to other students.

Administrator

The administrator manages verification, monitoring, complaints, and system activities.

Rider Use Cases

The rider can perform the following activities:

- Register Account
- Verify Identity
- Login
- Search Rides
- View Ride Details
- Request Booking
- Make Payment
- Track Ride
- Chat with Driver
- Trigger SOS
- Submit Review
- Report Complaint

Driver Use Cases

The driver can perform the following activities:

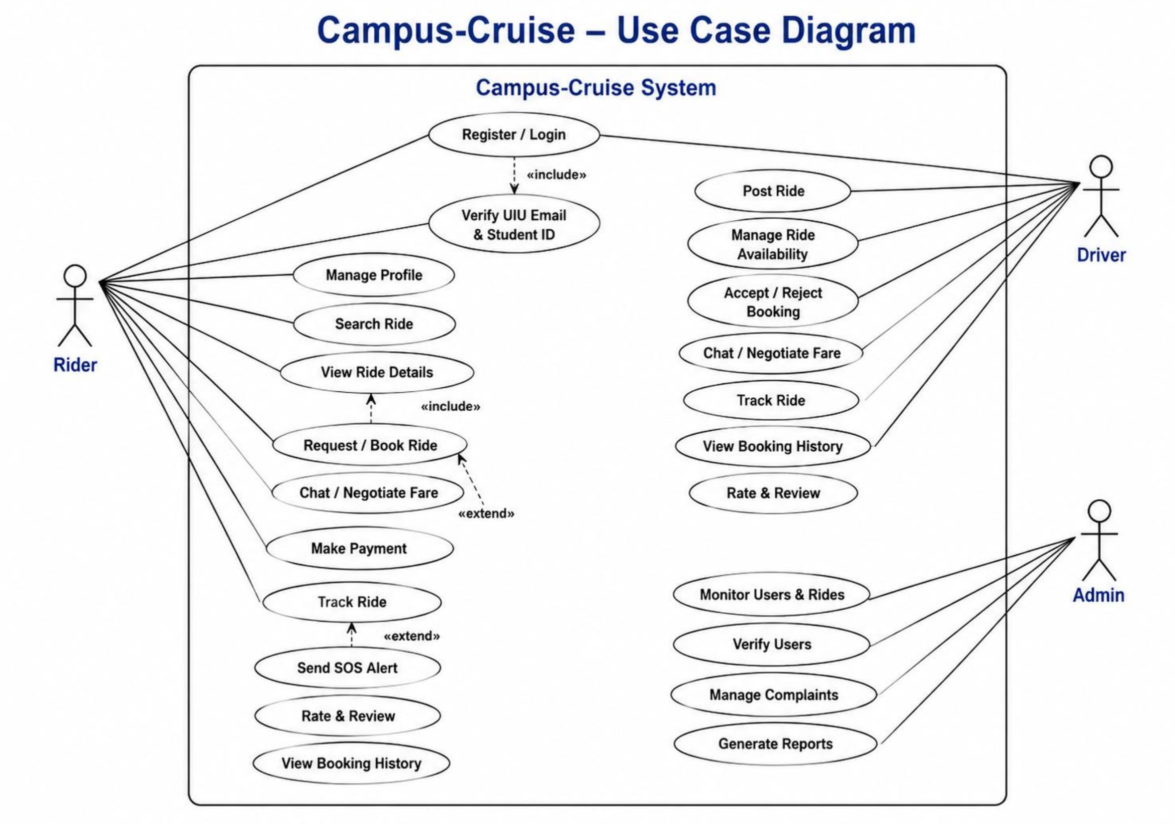
- Register Account
- Verify Driving License
- Login
- Create Ride
- Manage Ride
- Accept Booking Request
- Reject Booking Request
- Chat with Rider
- Receive Payment
- View Reviews

Admin Use Cases

The administrator can perform the following activities:

- Verify Users
- Verify Drivers
- Monitor Rides
- Manage Complaints
- Suspend Users
- Ban Users
- Generate Reports
- Monitor Payments

Figure 7.2: Use Case Diagram



7.4 Class Diagram

7.4.1 Overview

The Class Diagram represents the static structure of the Campus-Cruise system. It identifies the main classes, their attributes, methods, and relationships.

The class diagram serves as the foundation for database design and object-oriented implementation.

Main Classes

User

Represents all registered users.

Attributes

- UserID
- Name
- Email
- StudentID
- Password
- Role
- VerificationStatus
- Rating

Methods

- Register()
- Login()
- UpdateProfile()
- SubmitReview()

Driver

Represents users who offer rides.

Attributes

- LicenseNumber
- VehicleType
- VehicleNumber
- SeatCapacity

Methods

- CreateRide()
- ManageRide()
- AcceptBooking()
- RejectBooking()

Rider

Represents users who request rides.

Attributes

- PreferredRoute
- PaymentPreference

Methods

- SearchRide()
- BookRide()
- MakePayment()

Ride

Represents ride information.

Attributes

- RideID
- PickupLocation
- Destination
- DateTime
- Fare
- AvailableSeats

Methods

- CreateRide()
- UpdateRide()
- CancelRide()

Booking

Represents ride requests and confirmations.

Attributes

- BookingID
- RideID
- RiderID
- Status

Methods

- CreateBooking()
- ConfirmBooking()
- CancelBooking()

Payment

Represents payment transactions.

Attributes

- PaymentID
- Amount
- Method
- Status

Methods

- ProcessPayment()
- VerifyTransaction()

Complaint

Represents user complaints and reports.

Attributes

- ComplaintID
- Description
- Status

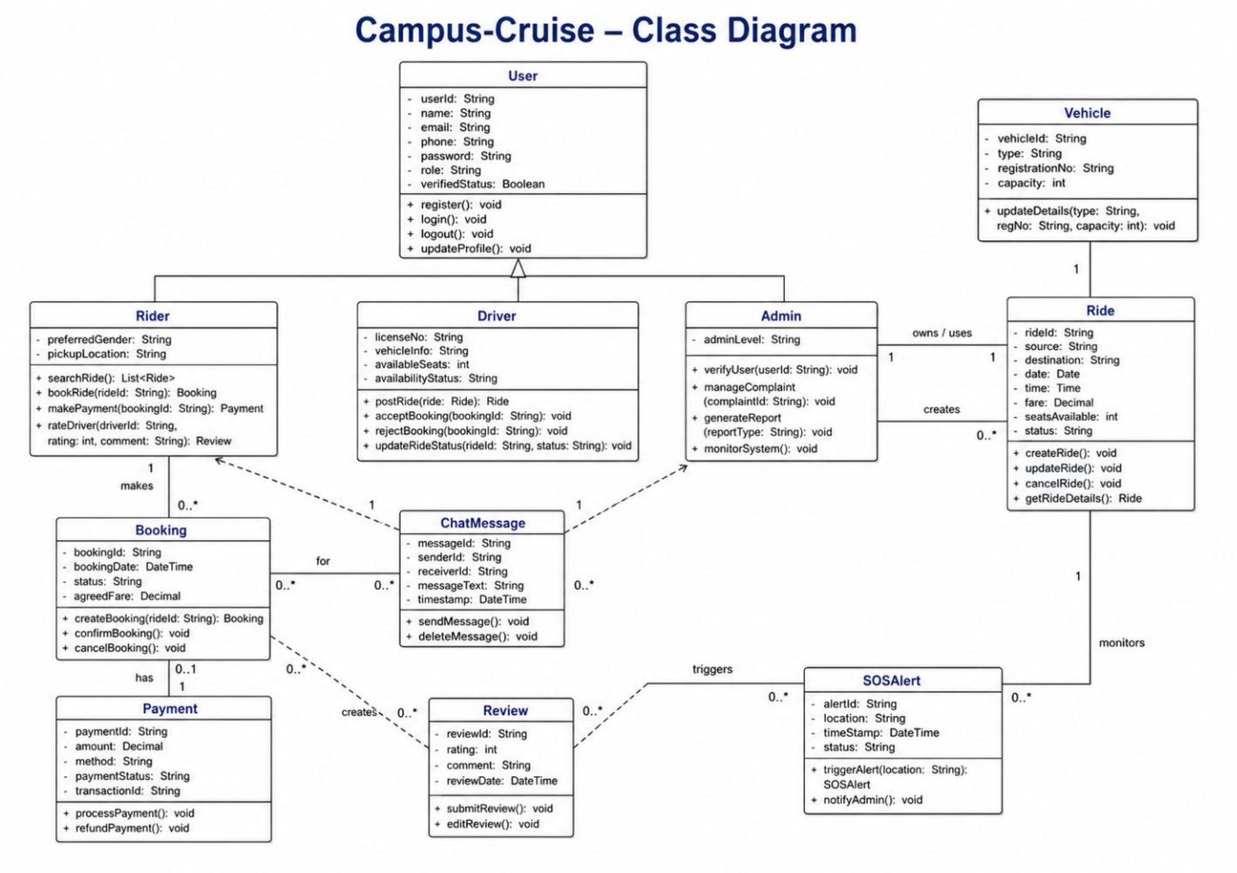
Methods

- SubmitComplaint()
- ResolveComplaint()

Relationships

- One Driver can create multiple Rides.
- One Ride can have multiple Bookings.
- One Rider can create multiple Bookings.
- One Booking can generate one Payment.
- One User can submit multiple Complaints.

Figure 7.3: Class Diagram



7.5 Sequence Diagram

7.5.1 Overview

The Sequence Diagram illustrates the order of interactions between users and system components during a specific process.

For Campus-Cruise, the ride booking process is considered the most important sequence.

Ride Booking Sequence

Step 1

The rider logs into the system.

Step 2

The rider searches for available rides.

Step 3

The system retrieves matching rides from the ride database.

Step 4

The rider selects a ride and submits a booking request.

Step 5

The system sends a notification to the driver.

Step 6

The driver reviews the request.

Step 7

The driver accepts or rejects the request.

Step 8

The system updates booking status.

Step 9

The rider receives confirmation.

Step 10

The rider and driver communicate through chat.

Step 11

The rider completes payment.

Step 12

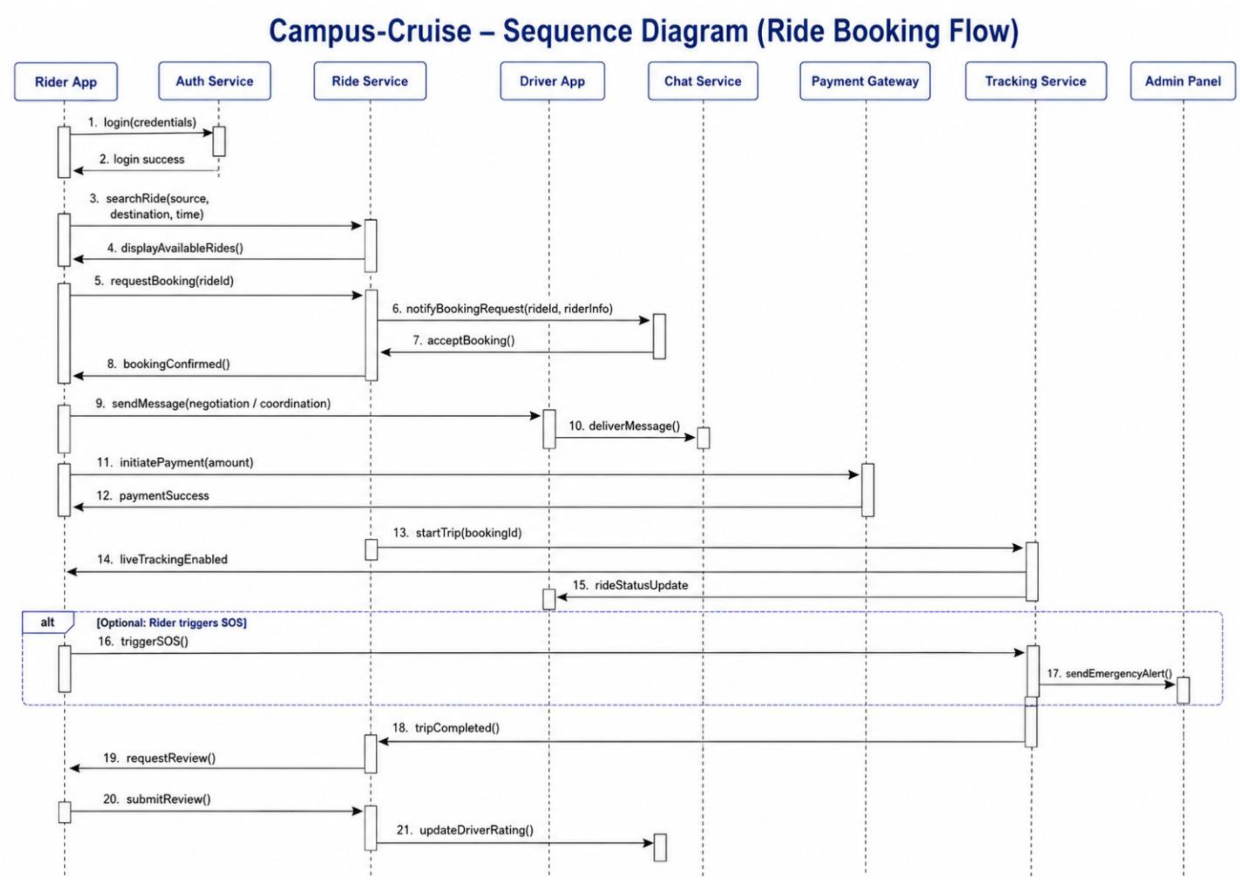
The ride begins and GPS tracking becomes active.

Step 13

The ride is completed.

Both users submit ratings and reviews.

Figure 7.4: Ride Booking Sequence Diagram



7.6 System Design Summary

The diagrams presented in this chapter provide a comprehensive view of the Campus-Cruise system. The DFD explains how information flows through the platform, the Use Case Diagram illustrates user interactions, the Class Diagram defines the system structure, and the Sequence Diagram demonstrates the execution flow of critical operations.

Together, these diagrams serve as a blueprint for implementation and provide a clear understanding of the overall system architecture and behavior.

8. Prototype Design

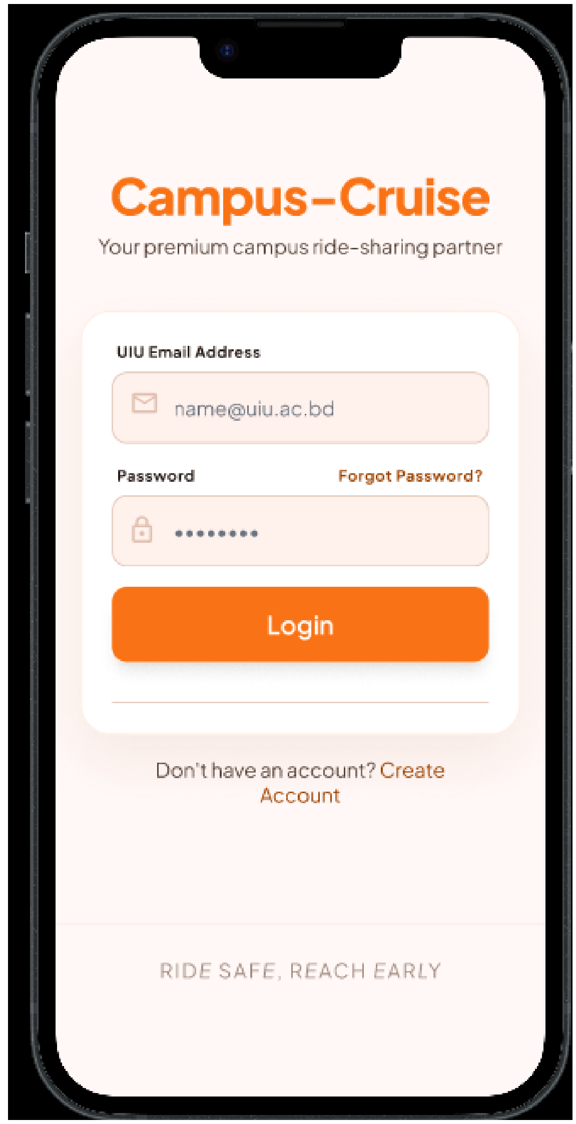
8.1 Introduction

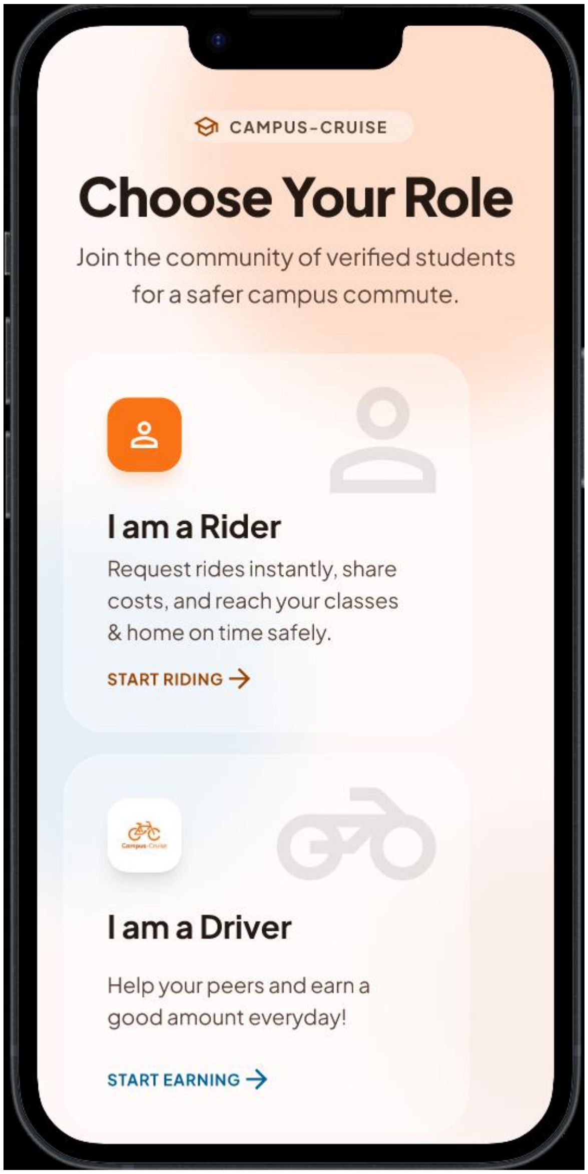
The Prototype Design phase focuses on creating a visual representation of the Campus-Cruise platform before full implementation. The prototype helps stakeholders, developers, and users understand the overall system workflow, navigation structure, and user experience. Both web and mobile interfaces were designed to ensure accessibility and usability for all UIU students.

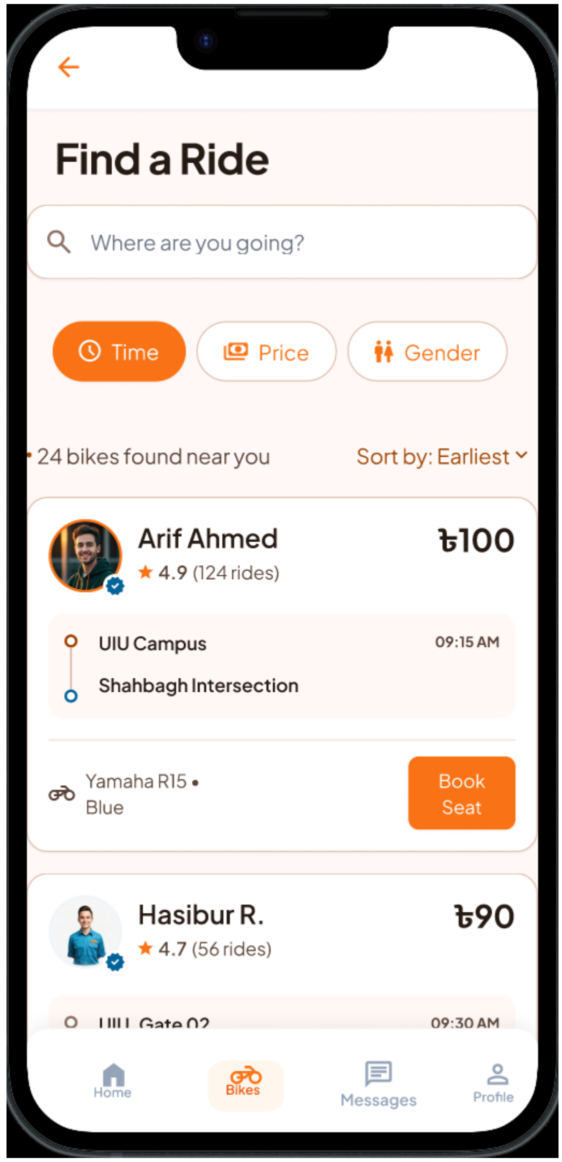
8.2 Mobile Application Prototypes:

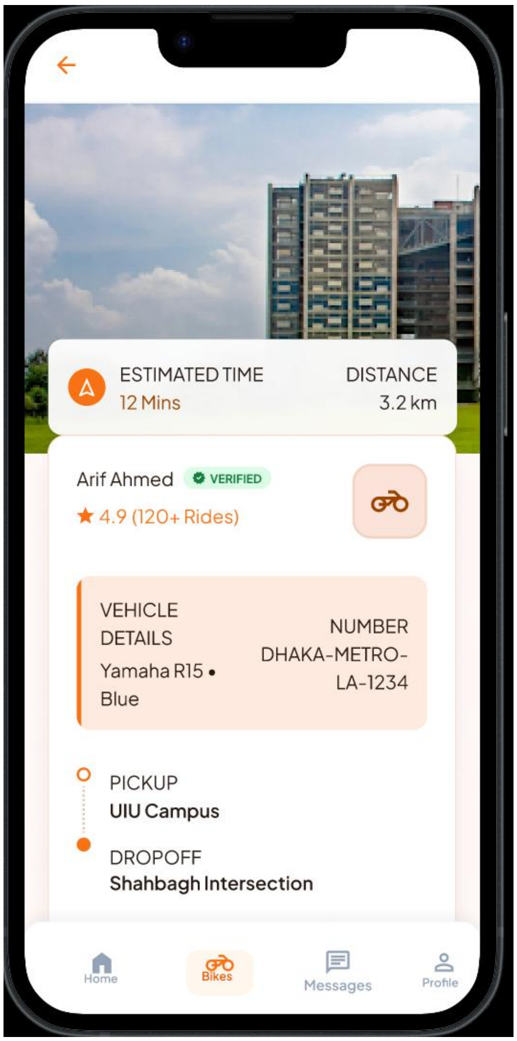
(Rider)


Campus-Cruise
Campus-Cruise
Safe • Affordable • Community
Fetching student drivers near your location

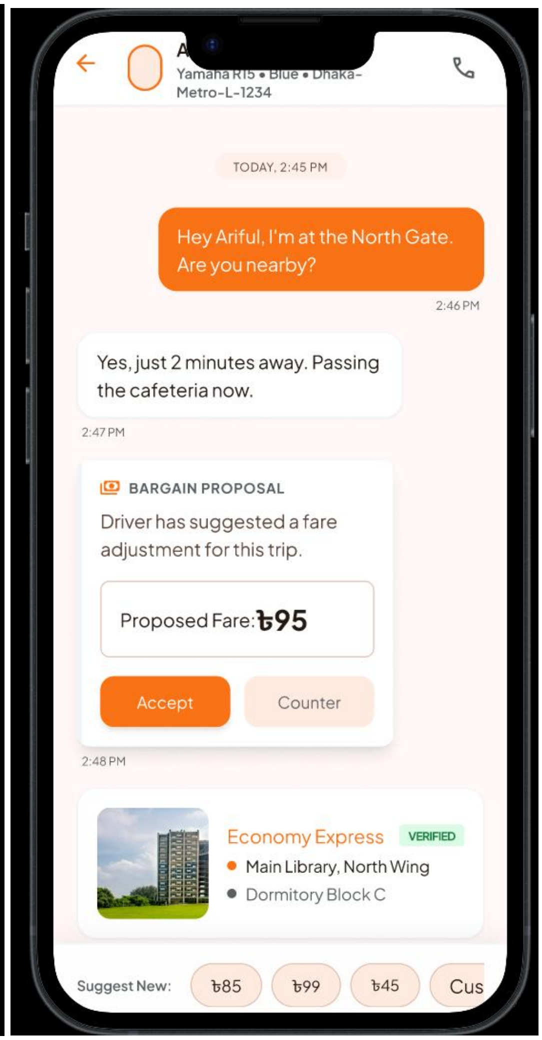

Campus-Cruise
Your premium campus ride-sharing partner
UIU Email Address
name@uiu.ac.bd
Password
Forgot Password?
Login
Don't have an account? Create Account
RIDE SAFE, REACH EARLY


CAMPUS-CRUISE
Choose Your Role
Join the community of verified students for a safer campus commute.
I am a Rider
Request rides instantly, share costs, and reach your classes & home on time safely.
START RIDING
I am a Driver
Help your peers and earn a good amount everyday!
START EARNING

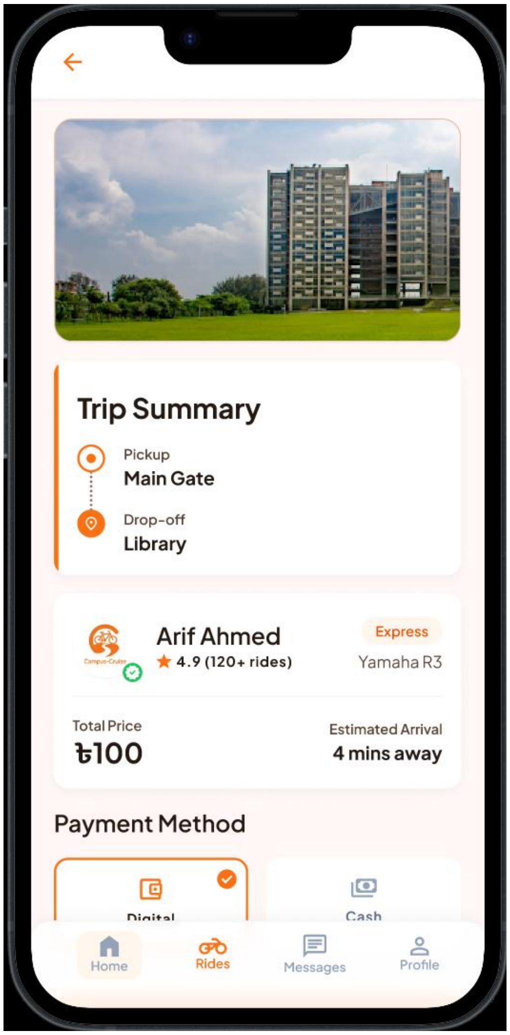

Find a Ride
Where are you going?
Time
Price
Gender
24 bikes found near you
Sort by: Earliest
Arif Ahmed
৳100
4.9 (124 rides)
UIU Campus
09:15 AM
Shahbagh Intersection
Yamaha R15 • Blue
Book Seat
Hasibur R.
৳90
4.7 (56 rides)
09:30 AM
Home
Bikes
Messages
Profile



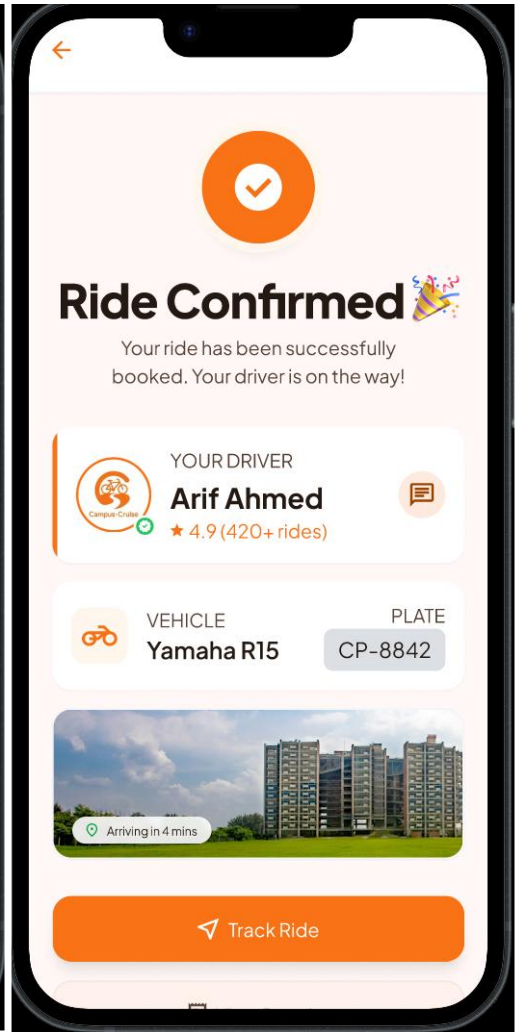
ESTIMATED TIME
12 Mins
DISTANCE
3.2 km
Arif Ahmed
VERIFIED
4.9 (120+ Rides)
VEHICLE DETAILS
Yamaha R15 • Blue
NUMBER
DHAKA-METRO-LA-1234
PICKUP
UIU Campus
DROPOFF
Shahbagh Intersection
Home
Bikes
Messages
Profile



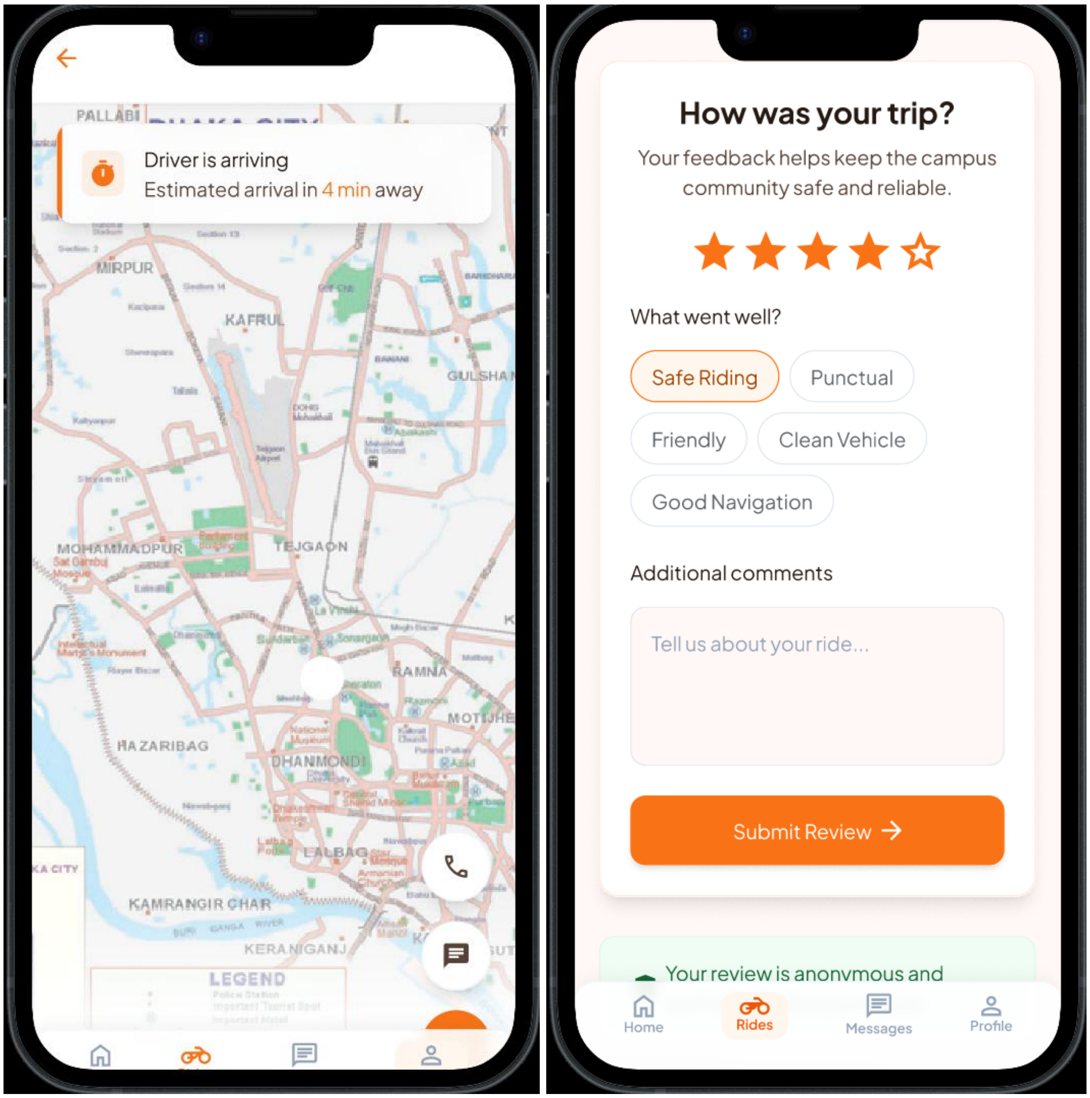
Yamaha R15 • Blue • Dhaka-Metro-L-1234
TODAY, 2:45 PM
Hey Ariful, I'm at the North Gate. Are you nearby?
2:46 PM
Yes, just 2 minutes away. Passing the cafeteria now.
2:47 PM
BARGAIN PROPOSAL
Driver has suggested a fare adjustment for this trip.
Proposed Fare: ৳95
Accept
Counter
2:48 PM
Economy Express
VERIFIED
Main Library, North Wing
Dormitory Block C
Suggest New:
৳85
৳99
৳45
Cus



Trip Summary
Pickup
Main Gate
Drop-off
Library
Arif Ahmed
Express
★ 4.9 (120+ rides)
Yamaha R3
Total Price
৳100
Estimated Arrival
4 mins away
Payment Method
Digital
Cash
Home
Rides
Messages
Profile



Ride Confirmed
Your ride has been successfully booked. Your driver is on the way!
YOUR DRIVER
Arif Ahmed
★ 4.9 (420+ rides)
VEHICLE
Yamaha R15
PLATE
CP-8842
Arriving in 4 mins
Track Ride


Driver is arriving
Estimated arrival in 4 min away
PALLABI
MIRPUR
KAFRUL
GULSHAN
MOHAMMADPUR
TEJGAON
RAMNA
MOTIJHEEL
HAZARIBAG
DHANMONDI
LALBAG
KAMRANGIR CHAR
KERANIGANJ
LEGEND
How was your trip?
Your feedback helps keep the campus community safe and reliable.
What went well?
Safe Riding
Punctual
Friendly
Clean Vehicle
Good Navigation
Additional comments
Tell us about your ride...
Submit Review
Your review is anonymous and
Home
Rides
Messages
Profile

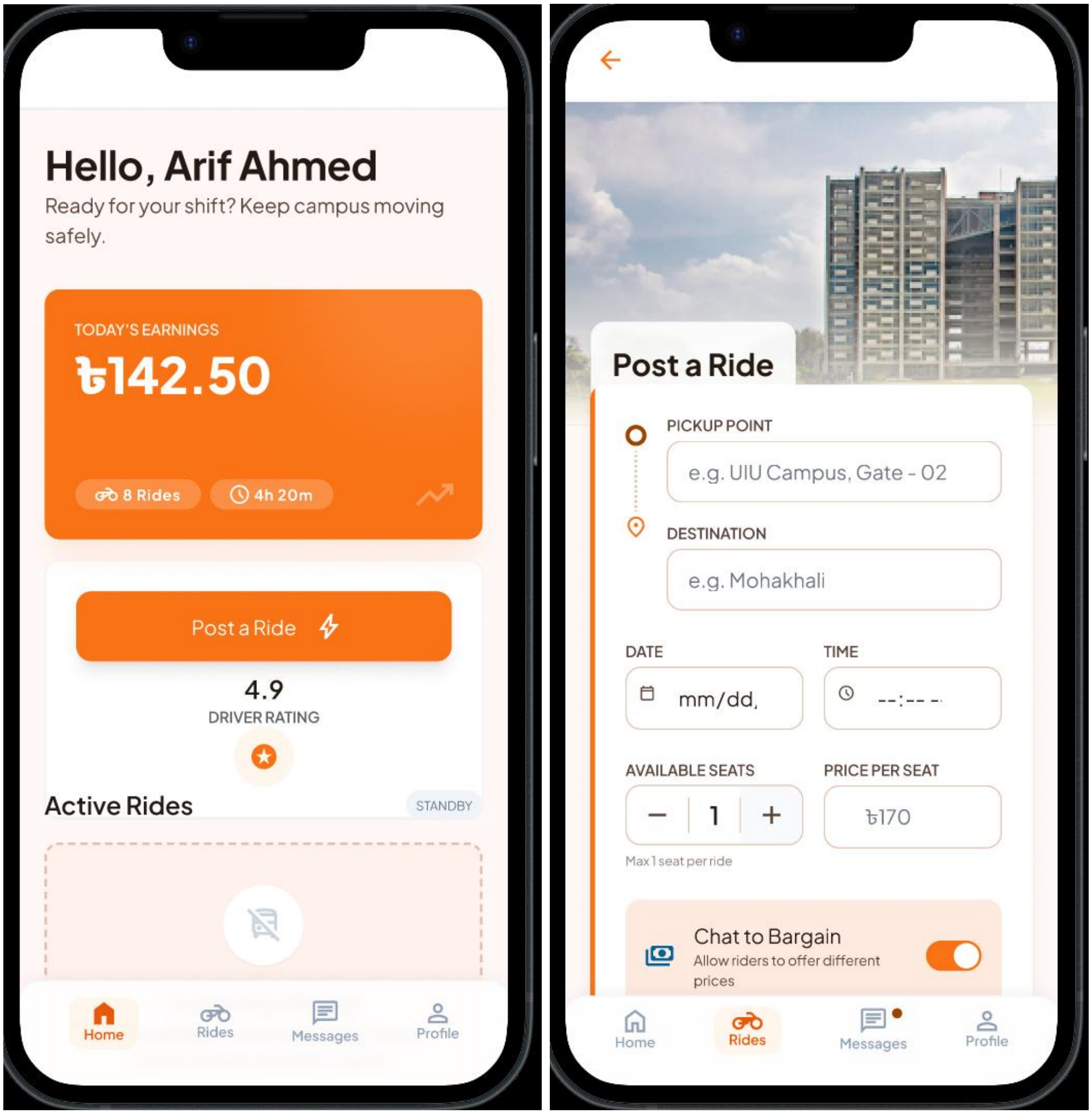
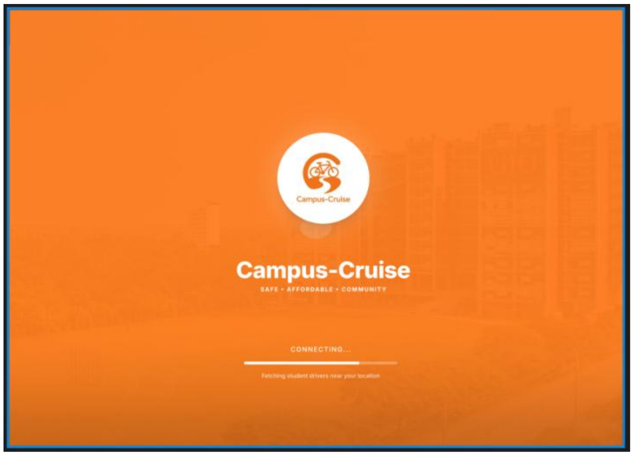
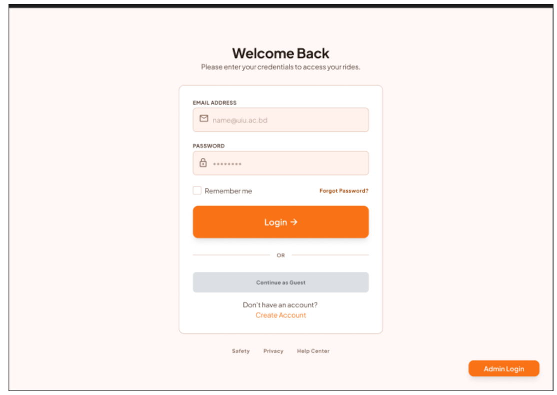


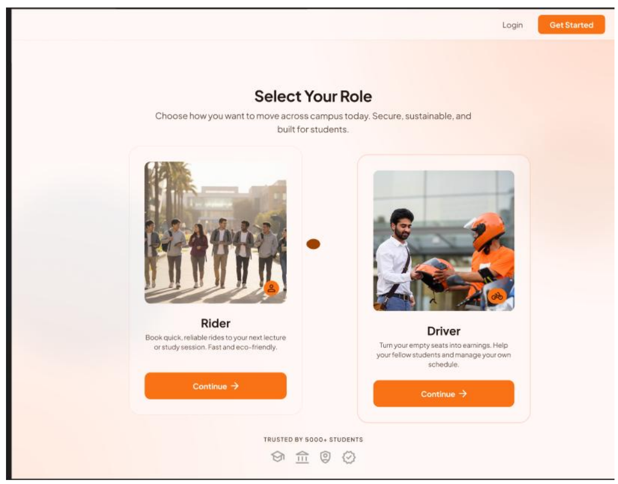
Figure: Driver UI

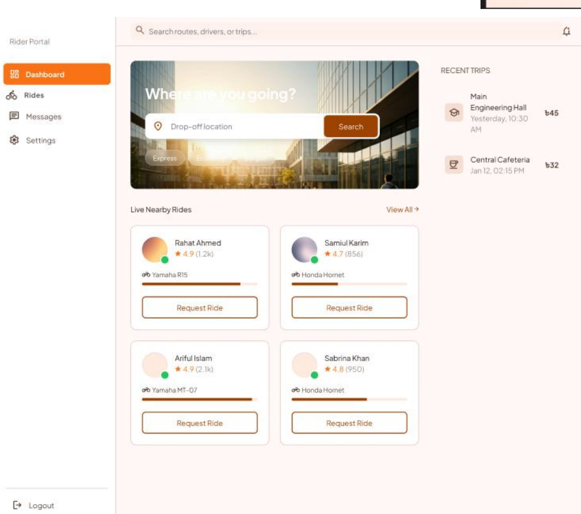


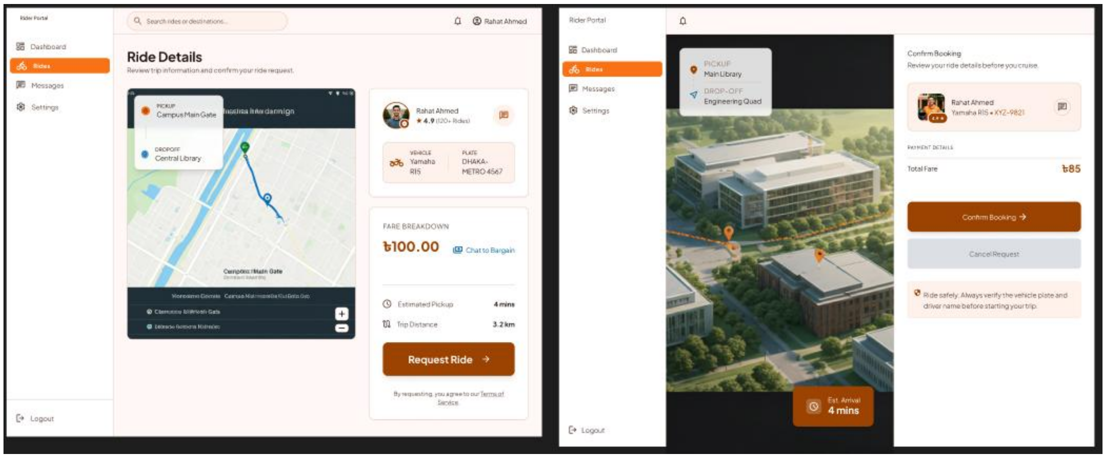
8.2 Web Application Prototypes:

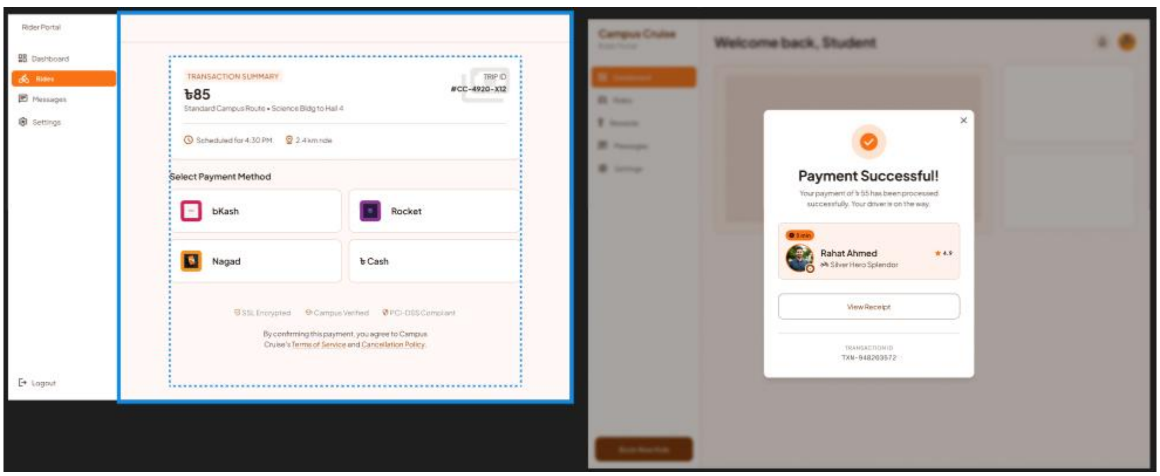
Figure: Rider Portal

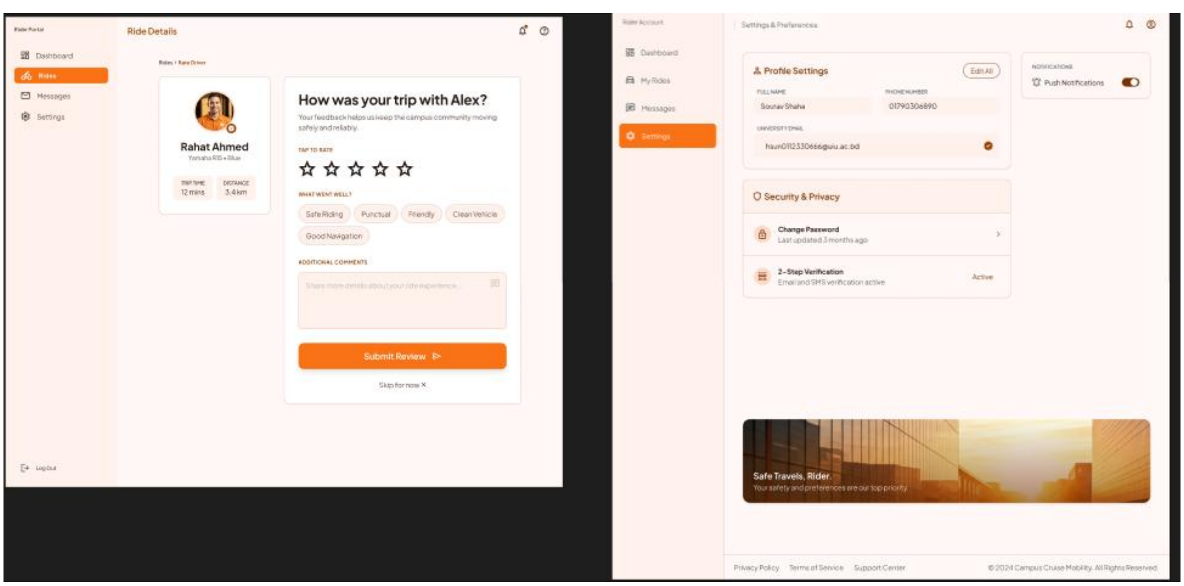







Dashboard
Rides
Messages
Settings
Logout
Rahat Ahmed
Ride Details
Review trip information and confirm your ride request.
PICKUP
Campus Main Gate
DROPOFF
Central Library
★ 4.9
FARE BREAKDOWN
৳100.00
Chat to Bargain
Estimated Pickup
4 mins
Trip Distance
3.2 km
Request Ride
Rider Portal
Main Library
DROP-OFF
Engineering Quad
Est. Arrival
4 mins
Confirm Booking
Review your ride details before you cruise.
Total Fare
৳85
Confirm Booking
Cancel Request
Ride safely. Always verify the vehicle plate and driver name before starting your trip.


Rider Portal
Dashboard
Rides
Messages
Settings
Logout
TRANSACTION SUMMARY
৳85
Standard Campus Route • Science Bldg to Hall 4
Scheduled for 4:30 PM
2.4 km ride
TRIP ID
#CC-4920-X12
Select Payment Method
bKash
Rocket
Nagad
৳ Cash
By confirming this payment, you agree to Campus Cruise's Terms of Service and Cancellation Policy.
Payment Successful!
Rahat Ahmed
View Receipt


Dashboard
Rides
Messages
Settings
Ride Details
Rahat Ahmed
How was your trip with Alex?
Your feedback helps us keep the campus community moving safely and reliably.
Safe Riding
Punctual
Friendly
Clean Vehicle
Good Navigation
Submit Review
Skip for now ×
Dashboard
My Rides
Messages
Settings
Profile Settings
Edit All
Push Notifications
Security & Privacy
Change Password
Last updated 3 months ago
2-Step Verification
Email and SMS verification active
Active
Safe Travels, Rider.
Privacy Policy
Terms of Service
Support Center

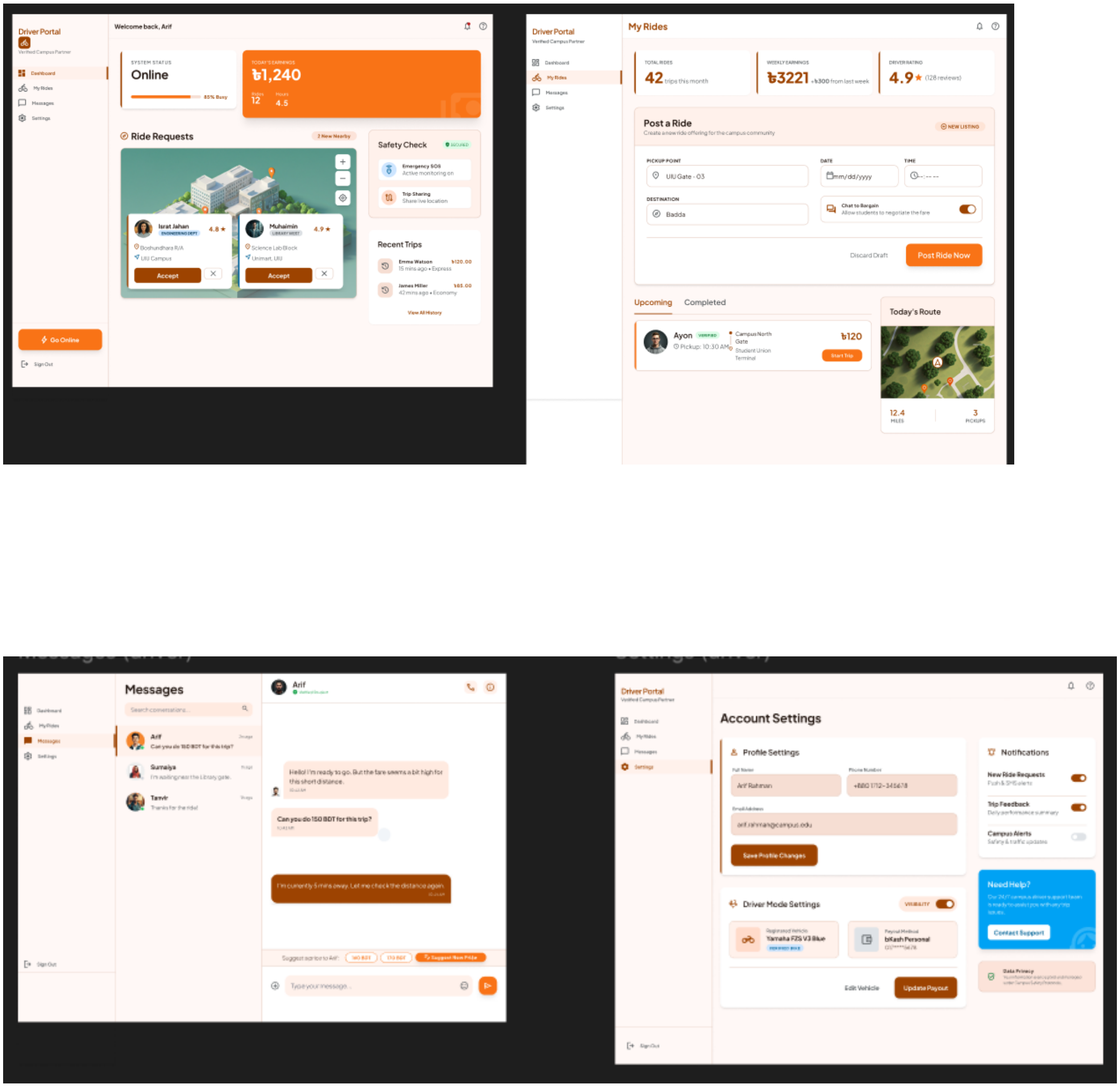


Figure: Driver Portal

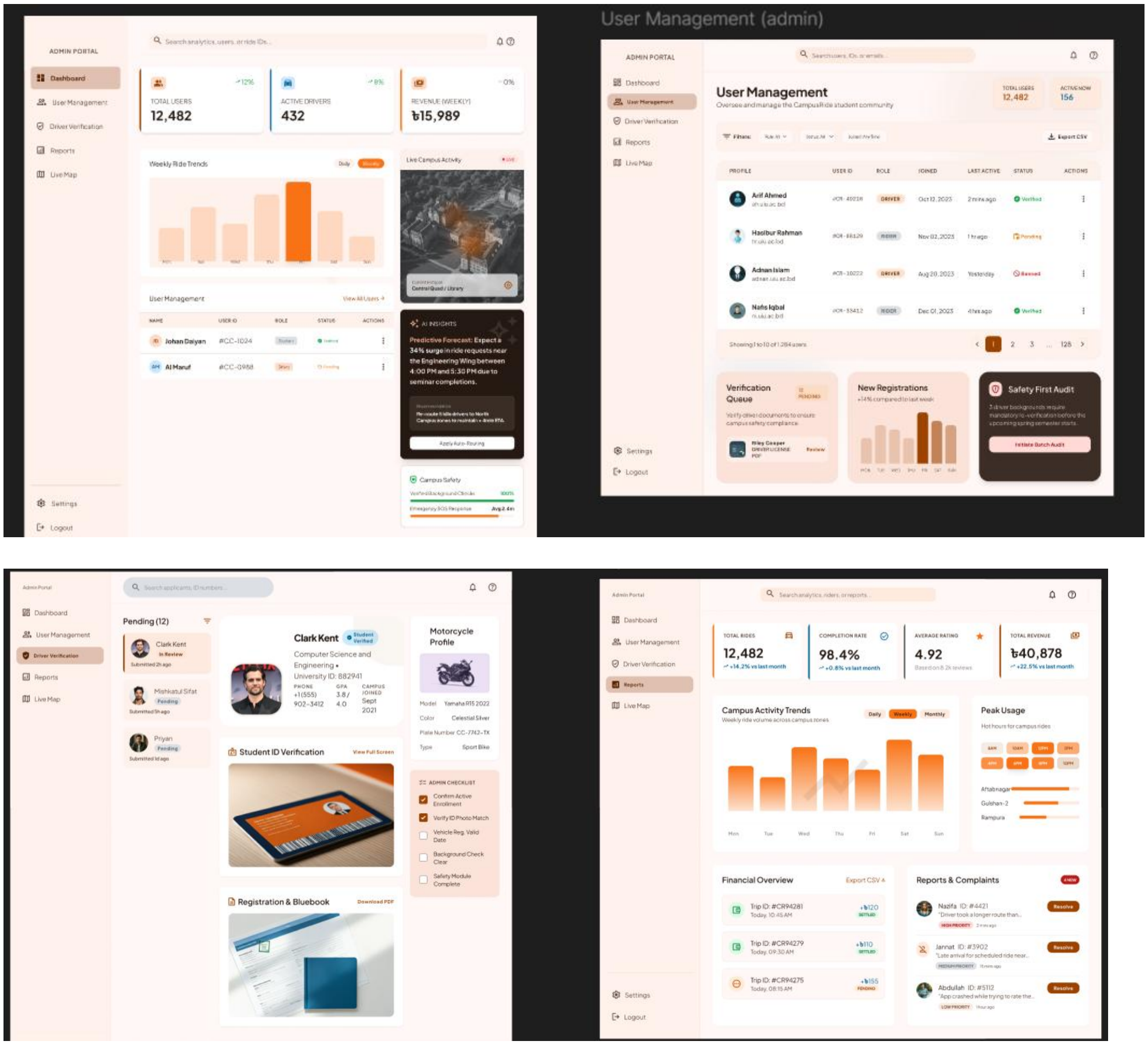


Figure: Admin Portal

8.3 UI Design Principles

The following UI/UX design principles were followed during the prototype development process:

1. Simplicity

The interface was designed with a clean and minimal layout to reduce user confusion and improve navigation.

2. Consistency

Consistent colors, fonts, buttons, and navigation patterns were maintained throughout the platform.

3. Accessibility

The design ensures easy usability for users with different technical skill levels and device types.

4. Responsiveness

The system is designed to function properly across mobile phones, tablets, and desktop devices.

5. User Feedback

Immediate feedback is provided for user actions such as login, booking requests, payments, and ride confirmations.

6. Visibility of System Status

Users can easily view ride status, booking status, payment status, and verification status at all times.

7. Error Prevention

Validation messages and confirmation dialogs help prevent user mistakes.

8. Mobile-First Design

Since students primarily use smartphones, the interface was designed using a mobile-first approach.

9. Conclusion

The Campus-Cruise project was developed to address the transportation challenges faced by students of United International University (UIU). Through requirement analysis, feasibility studies, benchmark comparisons, and system design activities, the project identified a practical solution for providing a secure, affordable, and efficient ride-sharing platform exclusively for verified UIU students.

The proposed system incorporates essential features such as student verification, ride posting, ride searching, booking management, real-time communication, GPS tracking, digital payment integration, emergency SOS support, and administrative monitoring. These features aim to reduce transportation costs, improve commuting convenience, enhance safety, and strengthen trust among students within the university community.

The feasibility analysis demonstrated that the project is technically, economically, operationally, and legally feasible. In addition, the system design models and prototype interfaces provide a clear foundation for future development and implementation. The platform is designed to create a reliable transportation-sharing ecosystem while ensuring user security, convenience, and accountability.

In conclusion, Campus-Cruise has strong potential to improve the daily commuting experience of UIU students by offering a trusted and community-driven ride-sharing solution. Future enhancements may include AI-based ride recommendations, predictive demand analysis, university shuttle integration, and expansion to other educational institutions. Through continuous improvement and user feedback, the platform can become a valuable transportation service for the university community.

10. References

Books

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2. Sommerville, I. (2016). *Software Engineering* (10th Edition). Pearson Education.
3. Dennis, A., Wixom, B. H., & Roth, R. M. (2020). *Systems Analysis and Design* (8th Edition). Wiley.

Websites

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3. Nagad Official Website. Available at: https://www.nagad.com.bd
4. Uber Official Website. Available at: https://www.uber.com
5. Pathao Official Website. Available at: https://www.pathao.com
6. Flutter Documentation. Available at: https://flutter.dev/docs

7. Node.js Documentation. Available at: https://nodejs.org/docs
8. Firebase Documentation. Available at: https://firebase.google.com/docs
9. W3Schools. Available at: https://www.w3schools.com
10. MDN Web Docs. Available at: https://developer.mozilla.org

Research and Survey Sources

1. Survey responses collected from UIU students using Google Forms.
2. Stakeholder discussions and team meetings conducted during requirements gathering and analysis.
3. Benchmark analysis of existing ride-sharing platforms including Uber and Pathao.
4. Online articles, blogs, and case studies related to ride-sharing systems, transportation management, and software engineering practices.

11. Appendices

Appendix A – Glossary

- **SRS** – Software Requirements Specification
- **UIU** – United International University
- **GPS** – Global Positioning System
- **API** – Application Programming Interface
- **UI** – User Interface
- **UX** – User Experience
- **DFD** – Data Flow Diagram
- **SDLC** – Software Development Life Cycle
- **SOS** – Emergency Support System

Appendix B – Additional Information

- Survey responses were collected from UIU students through Google Forms.
- Benchmark analysis was conducted using Uber and Pathao.
- System diagrams were prepared based on the identified requirements and functionalities.